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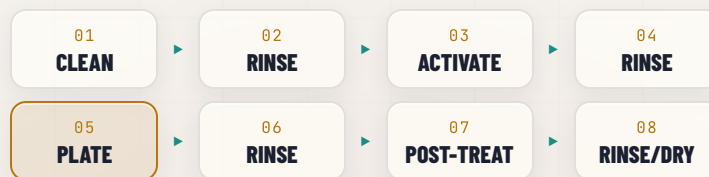
PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

SEMI-BRIGHT NICKEL

THE COMPLETE TRAINING MANUAL

DUPLEX BASE LAYER • LEVELED • DUCTILE • SULFUR-FREE

A full training course for the brand-new operator — from “what is an anode?” to a signed certificate of completion. Semi-bright nickel: the leveled, ductile, sulfur-free base layer of the duplex decorative stack. Assumes you know nothing. Teaches you everything — including the layer nobody sees.



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • BEGINNER-FRIENDLY STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, current Safety Data Sheets (SDS), and applicable OSHA, EPA, REACH, and local regulations before production use. Nickel compounds are classified as suspected carcinogens and are known skin sensitizers. **Regulatory data reviewed 2026-09-18:** exposure limits reflect published U.S. federal OSHA PELs as of this date; referenced standards should be checked against their current revision, and state limits (e.g., Cal/OSHA) and ACGIH TLVs may be stricter.

— FIND YOUR WAY

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **semi-bright nickel plating line**. Maybe you started this week. Maybe you have been racking parts on a decorative line for a year and finally want to know *why* there are two different nickel tanks that look almost identical. Either way, you are in the right place, and we are glad you are here.

Here is the most important thing we can tell you up front: **this manual assumes you know nothing about plating, chemistry, or electricity**. That is not an insult. That is a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what a “STEP test” is or why two nickel tanks that look the same must never, ever share a dosing cup. You learn it. We are going to teach it.

This manual is written in a **plain-language** style — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of hot acid nickel.

**REMEMBER**

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• A WORD BEFORE WE START: THE LAYER NOBODY SEES

Here is the strange, wonderful truth about the process you are learning: **nobody will ever look at what you plate**. On a finished part — a faucet, a bumper strip, a door handle — the semi-bright nickel layer is buried under a layer of bright nickel and a whisper of chrome. The customer sees the mirror. You build the thing *under* the mirror. And yet the semi-bright layer is the reason the part survives years of weather, road salt, and fingerprints instead of months. It is a **leveled, ductile, sulfur-free** nickel layer that is deliberately more *noble* than the bright nickel plated over it — which means, when corrosion finally arrives, the shiny layer sacrifices itself to protect yours. That trick is called the **duplex system**, it has its own chapter (Chapter 3), and it is the entire reason this bath exists. Keep this in the back of your mind the whole way through: **you are not making a finish. You are making the foundation that the finish stands on.**

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, rinse it, plate it, rinse it, complete the stack, and rinse/dry it. Reading it in order matters, because (as you'll soon learn) **the order of the plating line is not random** — and neither is the order of this book.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

**TIP**

A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a seasoned hand would lean over and tell you. These save you time and headaches.

**REMEMBER**

A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.

**HEADS-UP**

A safety warning. Read every single one. These are the notes that keep your skin, your eyes, and your lungs intact. We use them *liberally* because a nickel line has real hazards.

**TECHNICAL STUFF**

Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line perfectly. Read them and you will understand it on a deeper level.

**FUN FACT**

A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

**TIP**

Read the Fundamentals all the way through *before* you read any individual stage chapter. The stage chapters assume you already understand the big picture this section gives you — especially Chapter 3. If you read only one chapter in this book, read Chapter 3.

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Besides the margin icons, each station chapter ends with two recurring tools. The first is a **Teach-Back** checklist — the things you must be able to *say out loud, unprompted* before you're cleared on that station; it's how you and your trainer confirm the ideas actually stuck. The second is a bordered **Sign-Off** box at the very end of the station — the short statement your *trainer initials* (with you) once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU’LL BE ABLE TO DO

LEARNING OBJECTIVES

THE WHOLE FOUNDATION – MASTER THIS AND EVERY STAGE CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

- 1. **Explain what electroplating is** in plain language — how electricity moves metal from one place onto another.
- 2. **Describe what semi-bright nickel does** and why a shop runs it — a leveled, ductile, essentially sulfur-free satin deposit that serves as the corrosion-engineering base layer of the duplex decorative stack.
- 3. **Explain the duplex system** — why the semi-bright layer is more noble than the bright nickel above it, why that makes the bright layer corrode sacrificially and sideways, and how the ASTM B764 STEP test proves the stack will actually protect the part.
- 4. **Explain how semi-bright differs from BOTH of its siblings** — it levels (plain Watts doesn’t) and it is sulfur-free (bright nickel isn’t) — and why you must never dose bright-nickel brightener into a semi-bright tank.
- 5. **Name all eight stages of the line in order** and explain *why* the order cannot be rearranged.
- 6. **Understand what each tank is for** — what goes in, what comes out, and what “good” looks like at every handoff.
- 7. **Practice the firewall discipline** between the semi-bright and bright baths — dedicated rinses, tight drag-out control, and never sharing addition equipment between the two nickel chemistries.
- 8. **Recognize the main hazards** at each station — hot caustic, mineral acids, electrical, nickel mist and skin sensitization — and know the correct PPE and emergency response for each.
- 9. **Use the basic field tests** every operator relies on, like the water-break test, conductivity checks, and reading a Hull Cell or leveling panel for leveling and soundness (not brightness).
- 10. **Spot trouble early** — read a deposit and know whether the problem is upstream, in the bath, in the additive system, or in your handling.
- 11. **Talk the language** — use the right words so you can communicate clearly with your lead and your chemistry supplier.



REMEMBER

You are not expected to hit all of these on day one. Plating is a craft. The goal is steady competence, not instant mastery.

THE EIGHT-STAGE PROCESS AT A GLANCE

Here is the whole semi-bright line in one strip. Don’t memorize it yet — just get a feel for the rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Post-Treat → Rinse/Dry.**

#	STAGE	THE ONE THING IT DOES	TEMP	TIME
01	Alkaline Soak Clean / Electroclean	Removes all oil, grease, buffing compound, and shop soil	140–180°F	3–10 min
02	Rinse (Pre-Activation)	Washes off cleaner before the acid dip	Ambient	30–60 s
03	Acid Activation (Acid Dip)	Dissolves surface oxide; “wakes up” the metal	Amb–110°F	15–60 s
04	Rinse (Pre-Plate)	Guards the nickel bath from acid, iron & metals	Ambient	30–60 s
05	Semi-Bright Nickel Plating	Deposits the leveled, sulfur-free base layer	120–150°F	15–60+ min
06	Rinse (Post-Plate)	Recovers drag-out; keeps the two nickel chemistries apart	Ambient	30–60 s
07	Post-Treatment	Bright nickel + chrome complete the stack; bake if required	Varies	Varies
08	Final Rinse / Dry	Cleans up and dries without water spots	Amb–140°F	30–60 s + dry



REMEMBER

Notice the pattern: **a rinse follows almost every working tank.** Rinses are not breaks — they are checkpoints that keep one chemistry from poisoning the next. On a duplex line this matters *twice over*: the semi-bright bath is maintained rather than dumped (a well-run bath runs for years, so every impurity is a permanent guest), **and** the rinse after it is the only thing standing between two nickel chemistries that must never mix.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. Now let's unpack it so it actually means something.

Imagine you have a steel faucet body, a die-cast door handle, or a brass fitting that needs to look beautiful and survive years of wet, salty, fingerprinted service. You want to give it a tightly-bonded skin of **nickel** — a tough, corrosion-resistant, semi-noble metal — as the foundation for the layers that follow. You *could* try to spray or dip a coating on, but that would be uneven and weakly attached. Instead, we use electricity to grow a perfectly even, tightly-bonded layer of nickel directly onto the base metal, atom by atom. That process is electroplating.

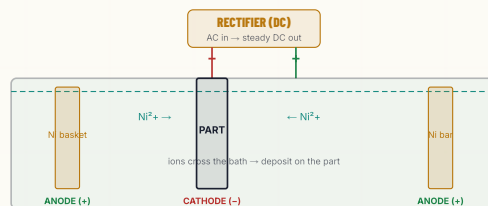
HOW ELECTRICITY MOVES METAL: THE KITCHEN-SINK PICTURE

Here is how it physically works. Picture a tank full of liquid called the **electrolyte** (or just “the bath”). This liquid has nickel dissolved in it — invisible nickel, broken down into tiny charged particles called **ions**. An *ion* is just an atom that carries an electrical charge. Dissolved nickel floats around the bath as positively-charged nickel ions, written **Ni²⁺**.

You hook two things up to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction “DC” current plating needs):

- **The cathode** — this is your *part*, connected to the **negative** terminal. Just remember “cathode = the part, the thing collecting metal.”
- **The anode** — bars or baskets of nickel hanging in the same tank, connected to the **positive** terminal. As the bath uses up its dissolved nickel, the anodes slowly dissolve to replace it — they are the “fuel tank” of nickel metal. (“Anode = positive = the metal that feeds the bath.”)

When you turn on the rectifier: (1) current flows, and the positively-charged nickel ions (Ni²⁺) are *attracted* to the negatively-charged part, because opposite charges attract; (2) when a nickel ion reaches the part's surface, it grabs electrons and turns back into solid nickel metal, sticking to the surface; (3) meanwhile, at the anodes, solid nickel dissolves *into* the bath, releasing new Ni²⁺ ions to keep the supply topped up. The net result: nickel moves off the anodes, through the bath, and onto your part. The longer you plate and the more current you push, the thicker the coating.



REMEMBER

Three words to lock in now: **Cathode = the part** (collects nickel). **Anode = the nickel** (feeds the bath). **Electrolyte = the bath** (carries nickel from one to the other). Almost everything in plating is a variation on this one idea.



TECHNICAL STUFF

At the part, a **reduction**: $\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}^0$. At the anode, the reverse — solid nickel gives up two electrons and dissolves: $\text{Ni}^0 \rightarrow \text{Ni}^{2+} + 2\text{e}^-$. Because the anodes replenish what the cathode consumes, a well-run nickel bath is largely self-balancing on nickel metal — that is one reason it can run for years.



FUN FACT

The core recipe under this bath — nickel sulfate, nickel chloride, boric acid — is the **Watts bath**, characterized by Oliver P. Watts in **1916**. More than a century later it is still the foundation under almost every nickel process in the world, including this one. Semi-bright nickel is a Watts bath *plus one clever additive* — and that one additive changes everything about the job the deposit does.



HEADS-UP

Plating rectifiers push enormous current — hundreds to thousands of amps, low voltage but very high current, into a tank full of hot conductive liquid. **Never reach into an energized tank, and never do maintenance until equipment is locked and tagged out (LOTO).** Electricity plus hot salt solution plus your hand demands total respect.

CHAPTER 2

WHAT IS SEMI-BRIGHT NICKEL?

A WATTS BATH WITH A SULFUR-FREE LEVELER – THE SATIN, DUCTILE, MORE-NOBLE MIDDLE CHILD OF THE NICKEL FAMILY

NICKEL PLATING COMES IN FAMILIES

“Nickel plating” just means coating something with nickel. But there are *several different recipes* for the bath that does it — and by now the catalog covers three of them, all built on the same Watts foundation:

- **Plain Watts nickel** — the core bath with **no additive system** at all (just an anti-pit wetter). Matte to semi-lustrous, ductile, sulfur-free, and it does **not** level. Engineering work: build-up, salvage, electroforming, undercoats. (*That’s the Watts Nickel manual — its own book in this series.*)
- **Bright nickel** — the core bath **plus a full brightener package**: a sulfur-bearing Class I carrier, a Class II primary brightener, and levelers. Mirror-bright, strongly leveling — and it deliberately co-deposits a little **sulfur**, which makes the deposit electrochemically *active*. (*That’s the Bright Nickel manual and poster series.*)
- **Semi-bright nickel** — the core bath **plus a sulfur-free leveler** and an anti-pit wetter, and *nothing else*. It **levels** like a decorative bath but stays **essentially sulfur-free (<0.005% S)** like an engineering bath. The finish is **satin / semi-lustrous** — deliberately not a mirror. **This manual is about semi-bright nickel.**



REMEMBER — SEMI-BRIGHT IS NEITHER PLAIN WATTS NOR BRIGHT NICKEL

1. **It is NOT mirror-bright** — the finish is satin / semi-lustrous, by design. Full brightness comes from the *bright* nickel layer plated over it.
2. **It DOES level** — unlike plain Watts, the sulfur-free leveler fills scratches, polish lines, and minor tool marks.
3. **It is sulfur-FREE (<0.005% S)** — no sulfur-bearing Class I carrier, no Class II brightener. This is what makes it more noble than the bright layer.
4. **It is NOT the top coat** — it is the *bottom* layer of a duplex stack: semi-bright → bright → chrome.
5. **It is deliberately MORE NOBLE than the bright nickel above it** — so the bright layer corrodes sacrificially and protects it. This is measured, not assumed (ASTM B764 STEP test — Chapter 3).
6. **It is ductile and low-to-moderate stress** — bends and forms without cracking; a sound base for the layers above.



HEADS-UP — THE OPERATOR GUARD

Do NOT dose bright-nickel (sulfur-bearing) brightener into a semi-bright tank — ever. A single drag-over of bright-bath additive can add sulfur to the semi-bright deposit, collapse the potential difference between the layers, and kill the corrosion performance of the whole stack — invisibly. The parts will look fine. They will fail in the field. If someone hands you a bottle from the bright line and tells you to “help the leveling along,” stop and ask your lead. This warning gets a whole discipline of its own in Chapter 6.

THE THREE NICKEL FINISHES, SIDE BY SIDE



PLAIN WATTS • MATTE • NOT LEVELED



SEMI-BRIGHT • SATIN / LEVELED (THIS MANUAL)



BRIGHT NICKEL • FULL MIRROR BRIGHT

Same Watts chassis under all three — the additive system (none / sulfur-free leveler / full brightener package) is what separates them. Satin, in the middle, is what success looks like on this line.

WHAT'S IN THE BATH

COMPONENT	TYPICAL RANGE	WHAT IT DOES
Nickel Sulfate (NiSO ₄ ·6H ₂ O)	240–330 g/L (32–44 oz/gal)	The main metal source — most of your nickel
Nickel Chloride (NiCl ₂ ·6H ₂ O)	30–45 g/L (4–6 oz/gal)	Helps anodes dissolve; carries current; improves throwing power — often kept toward the LOW end vs bright baths (see Ch 6)
Boric Acid (H ₃ BO ₃)	30–45 g/L (4–6 oz/gal)	A pH “buffer” — keeps acidity steady right at the part’s surface; prevents burning
Sulfur-free leveler (coumarin or coumarin-free)	Per supplier’s TDS; dosed on amp-hours	Leveling — WITHOUT sulfur co-deposit; the defining additive of this bath
Wetting / anti-pit agent	Per supplier’s TDS (low level)	Prevents gas pitting — this is NOT a brightener
pH	3.5–4.5	Lowered with H ₂ SO ₄ , raised with nickel carbonate — never NaOH



TECHNICAL STUFF

“g/L” means **grams per liter** — how many grams of a substance are dissolved in each liter of bath. “oz/gal” is the same idea in ounces per gallon, which many US shops still use. A “**TDS**” is a **Technical Data Sheet** — the instruction document your chemistry supplier provides for each product. When this manual says “per TDS,” it means *follow the supplier’s sheet*. Never guess at additive levels. And “**amp-hours (Ah)**” is current multiplied by time — the true measure of how much *work* the bath has done, and therefore how much leveler it has consumed. You’ll meet amp-hour dosing properly in Chapter 6.



REMEMBER — THE DEFINING ADDITIVE

The additive that defines a semi-bright bath is a **sulfur-free leveler** — traditionally **coumarin**, increasingly a **coumarin-free** proprietary leveler. There is NO sulfur-bearing Class I carrier and NO Class II primary brightener — that package belongs to the *bright* nickel bath. A low anti-pit wetter is the only other organic. Getting sulfur into this deposit — from the wrong additive, from bright-bath drag-over, or from anode residues where a truly sulfur-free system is specified — defeats the entire purpose of the layer.



REMEMBER — BORIC ACID IS NOT OPTIONAL

It looks like a minor ingredient, but it is the buffer that holds the right acidity in the thin film of liquid right against the part where plating actually happens. Neglect the boric acid and you get **burning** at high-current areas even when everything else looks fine. Keep it above 30 g/L — and note that it dissolves slowly, so add it with heat and patience.

• WHY DO SHOPS USE SEMI-BRIGHT NICKEL?

Because a single layer of bright nickel — however beautiful — cannot survive serious corrosion service on its own. Here is what semi-bright is good at:

APPLICATION	WHY SEMI-BRIGHT NICKEL
Duplex / multilayer decorative systems	The more-noble, sulfur-free base layer beneath bright nickel and chrome — the primary reason semi-bright exists
Automotive exterior trim & hardware	Long-life corrosion systems where a single bright layer will not pass salt-spray / CASS testing
Plumbing fixtures, appliance & marine hardware	High-corrosion service under decorative chrome
Leveling of imperfect substrates	The sulfur-free leveler fills scratches and polish lines a plain Watts bath would reproduce
Corrosion-engineering underlayer	Dense, ductile, sulfur-free — a durable barrier that stays protected by the layers above
Where sulfur-free is required under thermal service	No co-deposited sulfur to embrittle at temperature, unlike bright nickel



FUN FACT

The next time you walk through a parking lot, look at the chrome trim on an older, well-kept vehicle. If the shine has gone slightly cloudy in patches but hasn't blistered or rusted through, you may be looking at a duplex nickel system doing exactly what it was designed to do: the bright layer quietly taking the corrosion hit, sideways, so the base layer — and the steel — stay untouched. The best plating job on the car is the one you can't see.

• THE DEPOSIT — WHAT “GOOD” LOOKS LIKE

PROPERTY	SEMI-BRIGHT (AS-PLATED, WELL-RUN)	BRIGHT NICKEL (TYPICAL)	PLAIN WATTS (ENGINEERING)
Appearance	Satin / semi-lustrous, leveled	Full mirror bright, leveled	Matte to semi-lustrous, NOT leveled
Leveling	Yes (sulfur-free leveler)	Yes (brightener system)	None
Sulfur content	Essentially nil (<0.005% S)	Present (Class I carrier)	Essentially nil (no additive)
Electrode potential	More noble (protected in the stack)	Less noble (corrodes sacrificially)	n/a — not used in duplex stacks
Ductility	High — sound base layer	Lower	High
Hardness	~200-400 HV (condition-dependent)	~350-550 HV	~150-250 HV
Role	Base layer of the duplex decorative stack	Top nickel layer under chrome	Standalone engineering deposit

Hardness figures are typical published ranges and depend heavily on bath conditions — treat them as engineering guideposts, not a spec guarantee. Always confirm against your supplier's TDS and the customer specification.

• THE HONEST TRADEOFFS

A good operator knows the weaknesses too:

- **Not decorative as-plated.** Satin, not mirror. It needs the bright-nickel and chrome layers over it to be a finished decorative surface. On this line, satin is success — the shine is someone else's tank.
- **The leveler is a living thing.** It is consumed with the work, it must be dosed on amp-hours, and it breaks down — its breakdown products cause dull gray, hard-to-brighten areas if you don't carbon-treat on schedule (Chapter 6).
- **Two nickel chemistries, one line.** Carryover of the sulfur-free leveler forward into the bright bath — and of sulfur-bearing bright additive *back* into the semi-bright bath — are the classic duplex-line failures. The firewall discipline (Chapter 6) exists because of this.
- **Sensitive to metallic contamination.** Copper, iron, zinc, and chromium sneaking into the bath cause dark, hazy, rough, or high-stress deposits — unchanged from the other nickel baths.
- **Runs hot.** Typically 120–150°F, so you pay to heat the tank.
- **Hydrogen embrittlement risk** on high-strength steel — a bake is required (covered in detail later).



TIP

If a part comes off this tank satin and someone calls it a “defect,” your first move is to explain the process — satin is correct for semi-bright, and the mirror comes from the *bright* tank downstream. What you *should* be checking on that part is **leveling** (are the polish lines filled?) — not shine.

CHAPTER 3

THE DUPLEX SYSTEM: THE LAYER NOBODY SEES

THE WHOLE REASON THIS BATH EXISTS — READ THIS CHAPTER TWICE

This is the centerpiece of the manual. Everything else — every rinse target, every additive rule, every firewall warning — exists in service of the idea on these pages. Take your time here.

• 3.1 – THE PROBLEM: CORROSION DIGS STRAIGHT DOWN

Start with the failure we're trying to prevent. Put a single layer of bright nickel on steel, cover it with decorative chrome, and put the part outdoors. The chrome layer is beautiful but *thin* — and it always has tiny pores and micro-cracks. Sooner or later, corrosion finds one of those openings and reaches the nickel underneath.

Now here's the killer: corrosion attacking a single nickel layer behaves like a drill. It concentrates at that one small opening and **pits straight down** — through the nickel, to the steel. Once it touches steel, you get rust-through: the blistered, brown-bleeding chrome you've seen on neglected trim. The part had plenty of nickel *on average*. It just had zero nickel at the bottom of one pit.

• 3.2 – THE TRICK: MAKE THE TOP LAYER SACRIFICE ITSELF

The duplex system defeats the drill with electrochemistry. Here is the whole idea in three sentences:

1. Plate a **semi-bright** layer first — leveled, ductile, and **sulfur-free**, which makes it electrochemically **noble** (reluctant to corrode).
2. Plate a **bright** layer over it — the mirror-maker, whose sulfur-bearing brightener chemistry makes it electrochemically **active** (less noble, *willing* to corrode).
3. When corrosion arrives through the chrome, it hits a choice: attack the noble layer below, or the active layer it's standing in. **It always takes the easier target.** The bright layer corrodes **preferentially and sideways** — spreading laterally like a puddle instead of drilling down like a bit.

The result: the corrosion spends years chewing slowly, laterally, through the sacrificial bright layer — while the semi-bright base underneath stays intact, and the steel under *that* stays dry. Same total nickel thickness as a single layer; dramatically longer life. **The stack outlasts the single layer not because it has more metal, but because it controls *where* the corrosion goes.**



REMEMBER

"The bright layer takes the hit so the base layer doesn't have to." In a duplex stack, the shiny layer everyone admires is the designated sacrifice, and the satin layer nobody sees is the protected foundation. When corrosion reaches the nickel, it spreads sideways through the bright layer instead of drilling down to the steel. That one sentence is the entire duplex system.

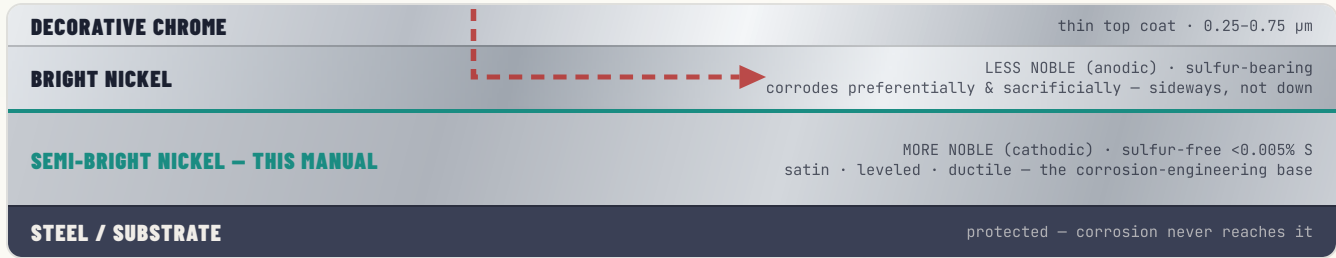


TECHNICAL STUFF — WHY SULFUR SETS THE POTENTIAL

The sulfur co-deposited by a bright bath's Class I carrier makes the bright nickel measurably more electrochemically **active** — its electrode potential sits *less noble* (more anodic) than sulfur-free nickel. Put the two layers in electrical contact (which they are — they're welded together atom-to-atom) in a corrosive electrolyte (rain + road salt will do), and you've built a tiny galvanic couple in which the bright layer is the anode. Anodes corrode; cathodes are protected. The semi-bright layer is the cathode of that couple. This is the same physics as a zinc anode protecting a boat hull — engineered into a decorative finish, at the scale of a few ten-thousandths of an inch.

THE DUPLEX STACK – WHY SEMI-BRIGHT EXISTS

CROSS-SECTION – REPRESENTATIVE, NOT TO SCALE • DASHED RED = THE CORROSION PATH



When corrosion reaches the nickel, the bright layer corrodes **preferentially and sideways** instead of pitting straight down to the steel — the entire reason a duplex system outperforms a single nickel layer of the same total thickness.

• 3.3 – NOBILITY IS A CHOICE YOU MAKE IN THE TANK

Here is the part that makes YOU central to the story. The potential difference between the layers is not automatic — it is **manufactured**, and it is fragile:

- The semi-bright layer is noble **only because it is sulfur-free** (<0.005% S). Every rule in this manual about keeping sulfur out — the right leveler, no bright-bath drag-over, the anode-selection care — exists to protect that number.
- The bright layer is sacrificial **only because it carries sulfur** from its Class I carrier. That's the bright line's job, and it's covered in the Bright Nickel manual.
- Contaminate the semi-bright layer with sulfur, and the two layers drift toward the *same* potential. The couple weakens or collapses. Corrosion stops choosing the bright layer — and goes back to being a drill.

And the cruel part: **a sulfur-contaminated semi-bright layer looks completely normal**. Satin finish, good leveling, passes every visual check. The failure is invisible until the part is in the field — or until somebody measures.



HEADS-UP

This is why the Chapter 2 Operator Guard is a *safety-of-product* rule, not a housekeeping preference: **never dose bright-nickel additive into the semi-bright tank, and never let the two chemistries share equipment**. You cannot see the damage. You can only prevent it — or measure it.

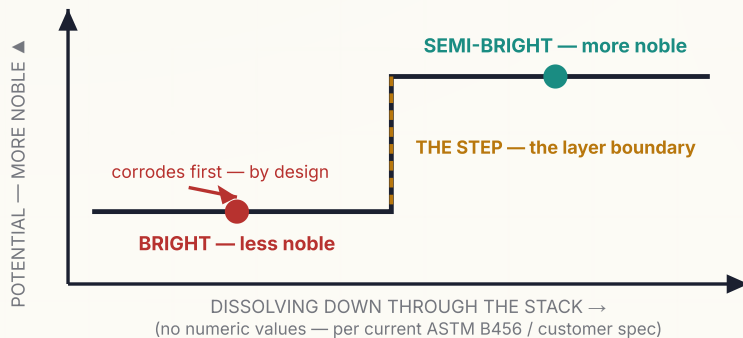
• 3.4 – THE STEP TEST: HOW WE PROVE THE STACK WORKS (ASTM B764)

So somebody measures. The industry's measuring stick is the **STEP test** — Simultaneous Thickness and Electrode Potential determination — standardized as **ASTM B764**.

Here's the plain-language version of what it does. A small cell is sealed against the finished part and the nickel layers are anodically dissolved down through the stack under controlled conditions, while the instrument records the electrode potential as each layer dissolves. Because the semi-bright and bright layers sit at different potentials, the recording shows a **step** — a jump in potential right at the boundary between the layers. From one run, you get:

- **The thickness of each nickel layer individually** (where the step lands tells you where the boundary is), and
- **The potential difference between the layers** (the height of the step tells you how strongly the bright layer will sacrifice itself for the base).

Right thicknesses but wrong potentials = a stack that will fail corrosion testing despite passing every thickness check. The STEP test is the only routine measurement that catches it.



STEP TEST — ASTM B764

LAYER ELECTRODE POTENTIAL — MORE NOBLE ▲

● **Semi-bright** — more noble (cathodic). Stays protected.

|| **THE POTENTIAL STEP — MEASURED, NOT ASSUMED**

● **Bright** — less noble (anodic). Corrodes first, by design.

One run yields each layer's thickness AND its potential. Right thicknesses, wrong potentials = a stack that fails corrosion testing.



REMEMBER

The required potential difference and the per-layer thickness split are set by **the current ASTM B456 and your customer's specification** — **not by this manual**. We deliberately print no numeric millivolt threshold: the number belongs to the spec on your desk, not a poster or a training book. What you must know cold is the *direction*: **bright must be less noble (anodic) to semi-bright**. If a STEP report shows that relationship weak or inverted, the stack is wrong no matter how good it looks.

• 3.5 – THE SPEC THAT GOVERNS THE STACK: ASTM B456

The decorative nickel-chromium system this manual serves — copper-plus-nickel-plus-chromium and nickel-plus-chromium coatings — is governed by **ASTM B456** (use the current revision). B456 defines the service-condition grades, the total nickel thickness, how that thickness splits between the layers, and the verification requirements the finished stack must meet; **ASTM B764** is its measuring stick for the layer potentials and thicknesses. Its international equivalent is **ISO 1456**. When a customer orders "duplex nickel per spec," B456 (or the customer's own document built on it) is what they mean.



HEADS-UP

Do not substitute the engineering-nickel specification (that belongs to the plain-Watts series) when quoting or verifying decorative duplex work — engineering nickel and decorative nickel-chromium are different specification systems with different requirements. When in doubt about which specification governs a job, ask your lead — don't guess from memory.

• 3.6 – BEYOND DUPLEX: TRIPLEX AND MICRODISCONTINUOUS CHROME

For the hardest service — think automotive exteriors in road-salt country — the industry goes one step further. A **triplex** system adds a very thin, high-activity nickel layer between the bright nickel and the chrome, and the chrome itself is plated **microporous or microcracked** on purpose. Thousands of engineered micro-sites spread the corrosion current over an enormous number of tiny anodic points, so no single site ever grows deep. Same philosophy as duplex — control *where* corrosion goes — pushed to its logical extreme. All of it lives under ASTM B456 and is verified by STEP and by accelerated corrosion tests (salt spray / CASS) per the spec.



FUN FACT

It sounds backwards to make chrome *deliberately porous* — but a chrome layer with thousands of microscopic openings outperforms a “perfect” one. A perfect chrome layer concentrates all the corrosion current at its few accidental flaws; a microporous one divides the same current across so many sites that none of them matters. In corrosion engineering, a million tiny doors beat one big crack.

• 3.7 – WHAT THIS MEANS FOR YOU, SHIFT TO SHIFT

You can't see nobility, and you won't run a STEP instrument on day one. So what does the duplex system actually ask of you?

1. **Keep sulfur out of your tank.** Right additive, dosed on amp-hours; no bright-bath bottles, cups, pumps, or drag-over; anode choice per the spec (Chapter 6).
2. **Respect the post-plate rinse** (Station 06) — it is the physical firewall between the two nickel chemistries.
3. **Chase leveling, not shine.** Your quality signal at this tank is *filled polish lines* and a sound satin surface, judged on a leveling panel — not brightness.
4. **Trust the sequence.** Semi-bright → bright → chrome, with the rinses between. Every handoff you do cleanly preserves a potential difference you'll never see.



REMEMBER

You've just learned the one idea that separates a duplex-line operator from someone who merely hangs parts: **corrosion protection here is built out of a potential difference, the potential difference is built out of sulfur control, and sulfur control is built out of your discipline.** Everything else in this book is detail.

CHAPTER 4

HOW A PLATING LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A LINE IS A SEQUENCE OF TANKS

A semi-bright nickel line is not one tank. It's a **sequence of tanks** that a part travels through in a strict order. Think of it like a car wash: each station does one job, and each job prepares the surface for the next. Parts ride down the line in one of two ways:

- **Rack plating** — Parts are individually hung on a **rack** (a frame with hooks or clips that holds parts and carries current to them). Used for larger, heavier, or delicate parts — and for most decorative work, where every visible surface matters.
- **Barrel plating** — Lots of small parts are dumped together into a perforated rotating **barrel** that tumbles them through the chemistry. Efficient for high volumes of small hardware, at lower current density.



TIP

You'll see different numbers for the same chemistry depending on rack vs. barrel — barrel baths often run a bit higher in nickel to make up for the lower current density tumbling parts receive. When in doubt about which numbers apply, check whether you're running rack or barrel and read the matching column on your shop-floor poster.

• THE WHOLE SEQUENCE: 8 STAGES

Here is the complete process. Don't memorize the details — just get a feel for the *flow*. Each stage gets its own full chapter later. The basic rhythm: **Clean → Rinse → Activate → Rinse → Plate → Rinse → Post-Treat → Rinse/Dry**.



• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS



REMEMBER

Every stage either protects the next stage or poisons it. There is no neutral step. A part that leaves the cleaner still carrying a film of cleaner will contaminate the acid. A part that leaves the acid still wet with acid — and dissolved iron — will contaminate the nickel bath. What you do (or fail to do) at Stage 1 shows up as a defect at Stage 5 — and on a duplex line, a defect at Stage 5 carries up through every layer above it. The line is a chain, and a chain is only as strong as its weakest link.

This is also why **the semi-bright bath itself is so precious**. Unlike the cleaner and the acid (dumped and remade regularly), a well-maintained semi-bright bath can run for *years*. But here's the catch: contaminants such as dissolved **iron, copper, and zinc** that sneak in tend to **stay in solution and accumulate**. You can't simply pour them out. Every milligram of junk you let drag into that tank is a permanent guest — you'll be dummy-plating and carbon-treating to chase it back out later.

And a duplex line adds one more system-level truth that a single-bath line never faces: **two nickel tanks that look nearly identical sit within a few stations of each other, and each one is poison to the other**. The semi-bright bath's leveler dulls the bright bath; the bright bath's sulfur-bearing additive destroys the semi-bright layer's nobility. The rinse between them isn't just a rinse — it's a border crossing.



TIP

Think of the semi-bright tank as the crown jewel of *this* line — and remember that the bright tank downstream is the crown jewel of *its* line. Everything before your tank exists to ensure only a perfectly clean, bare, well-rinsed part ever touches it. Everything after it exists to protect what you built — and to keep your chemistry and theirs from ever meeting.

CHAPTER 5

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here is the logic, link by link.

STAGE 1 — CLEAN: GET RID OF THE JUNK FIRST

Plating, acid, and every layer of the stack all need to touch *bare metal*. Oil, buffing compound, and shop soil block them. **Plate over dirt and you've just plated dirt** — the nickel bonds to grease, not metal, and peels off later. And remember what the leveler can and can't do: **semi-bright levels scratches, not dirt**. A cleaning failure here plates through every layer above it — bright nickel and chrome included. There is a free, foolproof test for cleanliness called the **water-break test**: if water sheets off in an unbroken film, the part is clean; if it beads up like on a waxed car, oil remains.

**HEADS-UP**

The cleaner is **hot (140–180°F)** and **strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Wear full PPE, and know where your shower and eyewash are before you need them.

STAGE 2 — RINSE (PRE-ACTIVATION): DON'T CARRY THE CLEANER FORWARD

The cleaner is alkaline (basic). The next tank is acid. If you carry cleaner straight into the acid, it neutralizes the acid, wastes it, and drags surfactants, silicates, and sodium downstream toward a bath where they don't belong. The rinse strips the cleaner film off first.

**REMEMBER**

"Drag cleaner into acid and you waste acid. Drag cleaner into the semi-bright bath and you poison the base layer of the whole stack."
Rinses exist to keep each chemistry from invading the next tank. The cleaner-to-acid rinse is your first firewall on the line.

STAGE 3 — ACID ACTIVATION: WAKE UP THE METAL

Even a clean surface grows an invisible layer of **oxide** (a tarnish-like film from reacting with air). Nickel can't bond to oxide. The acid dip dissolves that oxide and leaves a chemically "fresh," active surface for the nickel to grip.

- **Under-dip** (too short/weak) = oxide remains = nickel peels off later (adhesion failure). In a duplex system, adhesion of the semi-bright base carries the *entire stack* — if the bottom layer lets go, three layers fail as one.
- **Over-dip** (too long/strong/hot) = you start etching and pitting the part, and load it with hydrogen.

**REMEMBER**

No activation, no adhesion. Period. Activation has to come right before plating, with only a rinse in between — because freshly activated bare metal just grows a new oxide skin if it sits.

**HEADS-UP**

The acid dip uses **strong mineral acid** — typically HCl at 10–50% or H₂SO₄ at 5–20%. These cause severe burns and eye damage, and HCl gives off corrosive, choking fumes even at room temperature (OSHA PEL **5 ppm ceiling**). The acid eating into metal releases **flammable hydrogen gas**. Full acid PPE, good ventilation, and **always add acid to water, never water to acid**.

STAGE 4 – RINSE (PRE-PLATE): THE MOST IMPORTANT RINSE BEFORE THE TANK

This is the **guard rinse** protecting your most valuable tank. Acid dragged into the nickel bath lowers its pH, and — critically — carries dissolved **iron** (and any copper or zinc from the pickle) straight into a bath that is maintained for years. Those metals accumulate and are hard to remove — and iron in particular raises stress and cuts the ductility that makes semi-bright a sound base layer. This rinse targets a tighter conductivity (**< 200 µS/cm**) and double rinses are strongly recommended.



REMEMBER

"Every contaminant you allow through this rinse becomes a permanent guest in the base layer of every part you plate." Drain well, rinse hard, and move the part quickly to the nickel tank before fresh oxide can form.



TECHNICAL STUFF

The recurring villain that justifies all those rinses is **drag-out** (and its mirror image, **drag-in**). Drag-out is the film of solution that clings to a part as it leaves a tank; carried into the next tank it becomes drag-in — contamination. Operators measure rinse cleanliness with **conductivity**, in microsiemens per centimeter (µS/cm) — basically how many dissolved ions are in the water. *Higher conductivity = more dragged-in chemistry = a dirtier rinse.* Critical rinses on this line target **under 200 µS/cm**.

STAGE 5 – SEMI-BRIGHT NICKEL PLATING: THE MAIN EVENT

Now — and only now, with a perfectly clean, freshly activated, well-rinsed part — does the semi-bright layer go on. The bath is hot (120–150°F), acidic (pH 3.5–4.5), with the dissolving nickel anodes feeding fresh metal back into solution as you plate, and the sulfur-free leveler quietly filling the polish lines. It runs at high **cathode efficiency (93–97%)** — most of your current makes metal, not gas.

STAGE 6 – RINSE (POST-PLATE): RECOVER, FIREWALL, AND DECIDE ON THE BAKE

This rinse does *triple* duty on a duplex line. It recovers expensive nickel drag-out (a recovery rinse can capture it for return to the bath). It is the **firewall between the two nickel chemistries** — the barrier that keeps your leveler out of the bright bath and their sulfur-bearing additive out of yours. And it's the **decision point for the hydrogen-embrittlement bake** on hardened steel.

STAGE 7 – POST-TREATMENT: COMPLETE THE STACK

For most duplex work, "post-treatment" means the part goes on to the **bright nickel** tank and then to **decorative chrome** — completing the stack your layer anchors. For some jobs it means a bake, a seal, or a lacquer instead. Either way, the stack is then **verified** — thickness, and where specified, the STEP test.

STAGE 8 – FINAL RINSE / DRY: SEND IT OUT RIGHT

A last clean DI rinse and a warm (not scorching) dry finish the part without water spots or staining.



REMEMBER

The whole eight-stage sequence is one continuous logic chain: *clean it so nickel can stick → rinse so the cleaner doesn't ruin the acid → activate so the metal is bare and ready → rinse so acid and iron don't poison the precious nickel bath → plate the leveled, sulfur-free base layer → rinse to recover nickel and keep the two chemistries apart → complete the stack (bright + chrome) and bake if required → final rinse and gentle dry so it ships flawless.* Every step earns its place.



TIP

When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. A peeling stack? Look upstream at cleaning and activation. Dull gray patches the bright layer won't cover? Look at the leveler and its breakdown products. A duplex part that fails salt spray despite good thickness? Look at sulfur cross-contamination and the STEP report. The defect almost always points back to the stage that was supposed to prepare for it.

CHAPTER 6

THE SEMI-BRIGHT BATH: A DEEP DIVE

CHEMISTRY, WINDOWS, THE LEVELER SYSTEM, THE FIREWALL, ANODES, CONTAMINATION, STRESS, AND EMBRITTLEMENT

This chapter goes deeper than the others because the semi-bright bath is where the whole line pays off. Read it once now for the big picture; come back to it when you're troubleshooting.

• 6.1 – THE FIVE THINGS THAT ACTUALLY BUILD THIS DEPOSIT

On a bright nickel line, operators babysit a multi-part brightener system. On a plain Watts line, there are no additives to chase at all. Semi-bright sits between them: **four physical levers plus one additive** —

1. **Metal** (nickel concentration) — how much nickel is available to plate.
2. **pH** — the acidity, which controls stress, burning, and efficiency.
3. **Temperature** — higher temp gives lower stress and higher ductility; steady temp gives consistent leveling.
4. **Current density (CD)** — how hard you push; controls speed, and too much causes burning.
5. **The sulfur-free leveler** — dosed on amp-hours, monitored by leveling panel, protected from cross-contamination.



REMEMBER

"Four levers and one additive. The levers build the metal; the additive levels it; and keeping sulfur out of all five is what keeps the layer noble."
Master those and you master this bath.

• 6.2 – OPERATING WINDOWS (RACK VS. BARREL)

PARAMETER	RACK	BARREL	NOTES
Temperature	120–150°F (49–66°C)	120–150°F	Target 130–140°F; higher temp lowers stress; keep steady for consistent leveling
NiSO ₄ ·6H ₂ O	240–330 g/L	270–330 g/L	Higher for barrel to offset lower CD
NiCl ₂ ·6H ₂ O	30–45 g/L	40–52 g/L	Often kept LOW vs bright baths for lower stress; raise only enough for anode dissolution / throwing power
H ₂ BO ₃	30–45 g/L	37–45 g/L	Buffer — keep above 30 g/L; dissolves slowly
pH	3.5–4.5	3.5–4.5	H ₂ SO ₄ down, nickel carbonate up — never NaOH
Current Density	20–50 ASF (2–5.5 A/dm ²)	5–15 ASF (0.5–1.6 A/dm ²)	Higher CD = faster build but more burning/stress risk
Sulfur-free leveler	Per TDS — dosed on amp-hours	Per TDS	Coumarin OR coumarin-free; consumed and breaks down — track on Ah, NOT calendar time
Anti-pit wetter	Per TDS (low level)	Per TDS	Pit prevention only — NOT a brightener
Cathode Efficiency	93–97%	93–97%	Some H ₂ evolution always present
Agitation	Air + cathode-rod movement	Barrel 6–12 RPM	Steady agitation keeps leveling uniform and lowers pitting
Filtration	Continuous, 3–5 turnovers/hr	Continuous	5–10 µm filter bags; periodic carbon for organics/breakdown products
Anode:Cathode Ratio	2:1 to 1:1	Per basket config	Enough anode area to prevent passivation



TECHNICAL STUFF

"ASF" = **amps per square foot**, the current density unit most US shops use ("A/dm²" is the metric version). Current density is current spread over the area you're plating. Sharp edges and points see *higher* local CD than flat faces — which is why edges "burn" first if you push too hard.

 **TECHNICAL STUFF – THE CHLORIDE NOTE**
Many semi-bright formulations run chloride toward the **low end** of the nickel family (or use a low-chloride variant) to keep internal stress low and ductility high in the base layer. Chloride helps anodes dissolve and improves throwing power, but it also raises stress — and a base layer’s first job is to be *sound*. Confirm the target chloride against your supplier’s semi-bright TDS; it is often lower than the matching bright bath’s.

• **6.3 – THE LEVELER SYSTEM: COUMARIN AND ITS SUCCESSORS**

This is the additive that makes the bath what it is, so it deserves its own section — the way the brightener system gets one in the Bright Nickel manual, and the way *nothing* gets one in the Watts manual. (That contrast *is* the lesson.)

WHAT LEVELING ACTUALLY IS

A polished substrate still carries microscopic scratches and polish lines. A plating bath with **no** leveler (plain Watts) faithfully reproduces every one of them. A **leveler** is an organic additive that preferentially adsorbs on the tiny peaks of the surface, slowing deposition there, so the valleys fill faster than the peaks grow. Plate long enough and the surface comes out *smoother than it went in*. That’s leveling — and semi-bright does it **without sulfur**.

THE TWO CHEMISTRIES

APPROACH	CHEMISTRY	WHAT IT DOES	WATCH-OUT
Traditional	Coumarin (a sulfur-free leveler)	High leveling, ductile, sulfur-free semi-bright deposit over a wide CD range	Coumarin is consumed and breaks down ; its degradation products (melilotic-acid-type species and related compounds) build up and cause dull, gray, hard-to-brighten areas if not carbon-treated out
Modern	Coumarin-free proprietary levelers	Approach coumarin’s leveling while staying virtually sulfur-free; less troublesome breakdown	Follow the specific supplier TDS — dosing and carbon-treatment schedules differ from coumarin

DOSE ON AMP-HOURS, NOT ON THE CALENDAR

The leveler is consumed **by the work**, not by the clock. A busy Monday burns leveler; an idle Sunday doesn’t. So consumption is tracked in **amp-hours (Ah)** — current × time, totaled by the rectifier — and additions are made per your supplier’s TDS at set Ah intervals. Never dose “because it’s Tuesday,” and never dose by eye.

 **REMEMBER – THE TWO-SIDED LEVELER RULE**
Too little leveler and the bath stops leveling — polish lines reproduce, and the panel tells you before the parts do. **Too much breakdown product** and you get dull gray patches that the bright layer *cannot* cover. The fixes point opposite directions — *add* leveler per Ah for the first; *carbon-treat* for the second — which is why you confirm with a panel instead of guessing.

**TECHNICAL STUFF — WHERE THE COUMARIN GOES**

Coumarin doesn't just sit in solution; a portion is consumed into the deposit and a portion is electrochemically degraded at the electrodes. The degradation products — melilotic-acid-type species and related compounds — are not levelers. They accumulate, and past a point they show up as dull gray areas with poor response to the bright layer. That's why a coumarin bath lives on a **regular amp-hour carbon-treatment schedule** — you are cleaning up after the leveler's own metabolism. Modern coumarin-free products were developed largely to tame exactly this maintenance burden.

CONFIRMING THE LEVELER: PANEL, NOT SHINE

You verify the leveler with a **Hull Cell panel** or a **leveling panel** (a scribed or pre-scratched panel plated at a known CD): are the scratch lines filling? Is the satin uniform? You do **not** verify it by brightness — there is no brightener in this tank, and chasing shine here means someone has confused the two nickel baths (see the next section — that mistake has a name and a discipline built around preventing it).

• 6.4 — THE FIREWALL: SEMI-BRIGHT ↔ BRIGHT DISCIPLINE

Two nickel tanks. Nearly identical color, temperature, and smell. A few stations apart. **Each one is poison to the other.** This section is the discipline that keeps them apart — treat it with the same seriousness as a safety rule, because for the *product*, that's exactly what it is.

WHAT GOES WRONG, IN EACH DIRECTION

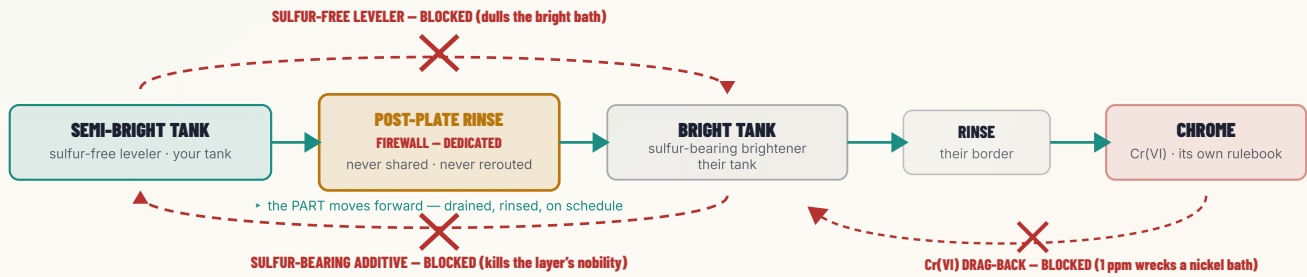
- **Bright → semi-bright (the killer):** sulfur-bearing bright additive dragged, splashed, or mis-dosed into the semi-bright tank puts **sulfur into the base layer**. The deposit still looks perfect. But its potential drifts toward the bright layer's, the STEP difference collapses, and the duplex mechanism dies — the stack fails corrosion testing (or worse, fails in the field) with every visual check passing.
- **Semi-bright → bright (the dulling):** sulfur-free leveler carried forward into the bright bath disrupts the brightener balance — dull, patchy, hard-to-control bright deposits, and hours of bright-bath troubleshooting for a problem that started in your rinse.

**HEADS-UP**

A firewall failure is invisible on the day it happens. No alarm sounds when sulfur enters the semi-bright tank. The parts plate normally, pass inspection, ship — and the failure surfaces weeks later as a collapsed STEP report or a field return. By then, the bath may need major treatment and every part since the event is suspect. This is why the rules on the next page are absolute, and why “it was just one cup” is the most expensive sentence on a duplex line.

THE FIREWALL — TWO NICKEL CHEMISTRIES, ONE BORDER CROSSING

SCHEMATIC OF THE DUPLEX MAIN LINE • CROSSED ARROWS = WHAT MUST NEVER CROSS



THE FIREWALL RULES

1. **Dedicated rinses between the two nickel tanks — always.** Never share a rinse tank between the semi-bright and bright stages. The post-plate rinse (Station 06) is a border crossing, not a formality.
2. **Never share addition equipment.** Pumps, dosing cups, measuring cylinders, transfer lines, filter carts — every piece of hardware belongs to ONE bath, is labeled for it, and never visits the other. A “clean-looking” cup can carry enough additive to matter.
3. **Tight drag-out control at the handoff.** Full drain time over the semi-bright tank, disciplined rack routing, and no shortcuts through the wrong rinse.
4. **Additions are deliberate acts.** Right bottle, right tank, right amount, per TDS, logged. If a container is unlabeled or in the wrong place, stop — quarantine it and ask. Never “probably.”
5. **Chrome stays out of everything.** Cr⁶⁺ drag-back from the downstream chrome tank is a live threat to BOTH nickel baths — as little as 1 ppm wrecks a nickel deposit. Rack routing and rinse discipline protect the whole nickel section from the chrome section.



TIP

Color-code it. Many duplex shops paint or tape semi-bright equipment one color and bright equipment another — cups, pump handles, lids, log binders. It turns the firewall from a memory exercise into something you can see from across the room. If your shop doesn't do this, suggest it — it's the cheapest insurance on the line.

• 6.5 — ANODE SELECTION: SULFUR-DEPOLARIZED VS. NON-SULFUR

This choice matters *more* for semi-bright than for either sibling, because the deposit must stay sulfur-free.

- **Sulfur-depolarized nickel (S-Rounds / S-nickel):** contains a trace of sulfur that keeps the anode dissolving cleanly and efficiently, even at higher current densities — the general workhorse anode. The trace of anode sulfur governs *anode dissolution* (residues stay behind in the bags as fines); it does not co-deposit the way a Class I carrier does. But where the spec demands a genuinely sulfur-free system, be deliberate about it.
- **Non-sulfur (rolled / electrolytic) nickel:** introduces no sulfur into the system — preferred where the semi-bright layer's sulfur content is tightly specified. It dissolves less readily and can **passivate** (go inert and stop dissolving) if anode area or chloride is marginal; if you run non-sulfur anodes, keep chloride adequate and anode area generous to prevent passivation.



TECHNICAL STUFF

"Sulfur-free deposit" for semi-bright refers to the absence of *co-deposited* sulfur from organic sulfur-bearing additives — that's the mechanism that sets the layer's potential. Anode sulfur is a different pathway with a much smaller footprint — but where the customer's sulfur-content requirement for the layer is tight, match the anode choice to it, and treat **anode residues where a sulfur-free system is specified** as part of your sulfur budget. Confirm against the customer spec; when in doubt, non-sulfur anodes with adequate chloride is the conservative call.



REMEMBER — ANODE BAGS

Always run **doubled polypropylene or cotton anode bags** to keep anode fines (little bits of undissolved anode) out of the bath — fines cause roughness. Leach new bags before first use.

• 6.6 – THE CONTAMINATION WATCH LIST

Because the bath is maintained for years, metallic contamination is your quiet, cumulative enemy. Know these action levels and what each contaminant does.

CONTAMINANT	ACTION LEVEL	EFFECT ON DEPOSIT	REMOVAL METHOD
Copper (Cu)	5 ppm	Dark streaks, roughness, poor low-CD deposit	Dummy plate at 2–5 ASF; carbon treat first
Iron (Fe)	25 ppm	Haze, roughness, raised stress, reduced ductility	Raise pH to 5.0–5.5 + H ₂ O ₂ to oxidize Fe ²⁺ →Fe ³⁺ ; filter; readjust pH
Zinc (Zn)	10 ppm	Hazy/dull, dark at low CD, adhesion loss	Dummy plate at 2–5 ASF; or raise pH to precipitate and filter
Chromium (Cr ⁶⁺)	1 ppm	Severe — dark/burnt, adhesion failure, skip plate	Reduce with sodium bisulfite to Cr ³⁺ , then treat as Cr ³⁺
Chromium (Cr ³⁺)	10 ppm	Dull, rough, poor throwing power and adhesion	Difficult — raise pH + dummy at low CD; or dilute
Lead (Pb)	0.5 ppm	Dark streaks, dendrites, stress cracking	Dummy plate at 2–5 ASF; check anode purity
Cadmium (Cd)	0.5 ppm	Peeling, blistering, dark deposits	Dummy plate; investigate source (legacy Cd drag-back)



REMEMBER – IRON HURTS YOU TWICE

Above ~25 ppm, iron doesn't just cause haze and roughness — it **raises internal stress and cuts ductility**, directly attacking the properties that make semi-bright a sound duplex base. Keep the pickle fresh and the pre-plate rinse tight; iron drag-in is the most common way iron gets in.



HEADS-UP – CHROMIUM IS THE WORST OFFENDER

Even **1 ppm** of hexavalent chromium (Cr⁶⁺) wrecks adhesion and causes burnt, skip-plated deposits. On a duplex line the chrome tank is *on your own line*, a few stations downstream — which makes Cr drag-back a live, daily threat instead of a theoretical one. Guard your rinses and rack routing fiercely; chrome never re-enters a nickel bath.

• 6.7 – READING A HULL CELL / LEVELING PANEL (LEVELING AND SOUNDNESS, NOT BRIGHTNESS)

A **Hull Cell** is a small angled test tank that plates a single panel across a whole *range* of current densities at once — high CD at one end, low CD at the other. On a bright nickel line, you’d read it for brightness response. **On a semi-bright line, you read it for leveling and soundness:** are the scratch lines filling, is the satin uniform, and is there burning, roughness, pitting, or contamination? Where bright nickel reads “dull vs bright,” semi-bright reads “leveled vs unleveled / gray.”

PATTERN (267 ML, 2A, 10 MIN, ~130-140°F)	LIKELY CAUSE	CORRECTIVE ACTION
Sound, leveled, uniform satin across panel	Bath in balance	Continue monitoring leveler (Ah), metals, pH, boric acid
Poor leveling — scratches reproduced	Leveler low or depleted	Add sulfur-free leveler per Ah; re-panel
Dull gray patches, not brightened by later bright layer	Coumarin breakdown products accumulating; organic (oil) drag-in	Carbon treat; resume leveler on Ah schedule
Burnt / powdery at high CD	pH too HIGH; boric acid or Ni metal low; temp too LOW; CD too high	Check boric acid, pH, Ni metal; raise temp; lower CD
Rough / nodular (sandpaper feel)	Particulate — poor filtration; anode fines	Check filter pressure; inspect/replace anode bags
Gas pitting across panel	Low anti-pit wetter; poor air agitation	Add wetter; check air supply and sparger
Hazy / dull at low CD	Zinc, copper, or lead contamination	Analyze metals; dummy plate at 2-5 ASF
Dark streaks/spots at low CD	Copper or lead contamination	Analyze Cu and Pb; dummy plate
Peeling when panel is bent	Adhesion failure (prep) OR raised stress from metals/iron	Recheck cleaning/activation; analyze metals; carbon treat



REMEMBER — THE BURNING DIRECTION, LOCKED IN

Burning at high CD is caused by pH too **HIGH** and temperature too **LOW** (plus low boric acid, low nickel metal, or CD too high). If burning appears, the moves are: check boric acid, *lower* the pH toward the window, *raise* the temperature toward the window, and back off the current. The inverted form of this rule shipped once on a sibling series and was corrected — do not trust anyone’s memory over this table.

• 6.8 – INTERNAL STRESS AND DUCTILITY

Every electrodeposit carries some **internal stress** — a built-in tension (or, less often, compression) locked into the metal as it grows. Too much *tensile* stress and the base layer can crack — and a cracked base layer is a failed stack, no matter how good the layers above are. Semi-bright is engineered for **low-to-moderate stress and high ductility**, and you hold it there with the same levers as any Watts-type bath:

- **Temperature** — higher temperature lowers stress and raises ductility. This is why the target is 130–140°F, not the low end.
- **pH** — lower pH tends to lower stress (but raises hydrogen evolution). It's a balance.
- **Current density** — pushing too high raises stress (and burning).
- **Chloride** — more chloride, more stress; part of why semi-bright baths often run chloride low (Section 6.2).
- **Contamination** — iron above the watch level raises stress and cuts ductility (Section 6.6).



TIP

If a stack cracks or the base layer checks during forming, don't reach for an additive. Look at **contamination (iron/metals), current density, chloride, and temperature** — the stress that cracked it was built in the tank.

• 6.9 – HYDROGEN EMBRITTLEMENT (AND WHY A DUCTILE DEPOSIT STILL NEEDS THE BAKE)

Here is a subtle but life-critical idea. During cleaning, acid activation, and plating, some **atomic hydrogen** gets absorbed into the **steel substrate** — not the nickel. On high-strength/hardened steel (≥ 40 HRC / ≥ 180 ksi UTS), trapped hydrogen can cause the part to **crack suddenly under load**, sometimes hours or days after it's in service. On a critical fastener or structural part, that's a real safety failure.



HEADS-UP – A DUCTILE DEPOSIT DOES NOT EXEMPT THE PART FROM THE BAKE

People get confused here: “semi-bright is ductile and low-stress, so no bake, right?” **Wrong.** The ductility is a property of the *nickel*. The embrittlement is in the *steel underneath*, from hydrogen absorbed during processing. **High-strength steel still requires a post-plate bake.**

THE RULE (PER ASTM B850)

- Substrates ≥ 40 HRC (≥ 180 ksi UTS) require baking.
- Bake at $375 \pm 25^\circ\text{F}$ for a **minimum of 8 hours**.
- The clock starts at completion of the **last electrodeposition step** — on a duplex/multilayer part that is usually the chrome or the bright nickel, so **verify per the customer spec** whether the bake comes after the full stack or at an intermediate point.
- **Bake within 4 hours** of that last plating step — no exceptions.
- Where required, embrittlement is verified by test per **ASTM F519**.

• 6.10 — KEEPING THE BATH ALIVE: PURIFICATION ON A SCHEDULE

Because you never dump it, you *maintain* it. Two routines keep a semi-bright bath healthy for years:

CONTINUOUS PURIFICATION

Filtration (3–5 turnovers/hr through 5–10 µm bags), plus **dummy plating** (plating onto a scrap cathode at low CD, 2–5 ASF, to pull metallic contaminants out) as needed.

PERIODIC CARBON TREATMENT — AND HERE SEMI-BRIGHT DIFFERS FROM BOTH SIBLINGS

It runs on a **regular amp-hour schedule** specifically to clear *leveler breakdown products*, not just incidental organics. (Plain Watts has no leveler to decompose, so it carbon-treats least often; bright nickel treats to clear brightener breakdown; semi-bright treats to clear leveler breakdown — same tool, three different reasons.)

1. Raise pH to 5.0–5.5 with nickel carbonate.
2. Add ~3% v/v H₂O₂ (30% grade) to oxidize Fe²⁺ → Fe³⁺.
3. Heat to 140–150°F.
4. Add 1–3 g/L powdered activated carbon; stir 2–4 hours.
5. Filter through a 1–5 µm filter to remove carbon, precipitated metals, and leveler breakdown products.
6. Readjust pH to 3.5–4.5 with dilute H₂SO₄.
7. Dummy plate at 2–5 ASF for 4–8 hours to remove residual metals.
8. **Re-dose the sulfur-free leveler per TDS** (carbon strips it along with the breakdown products), then run a leveling panel / Hull Cell to verify leveling before resuming production.



HEADS-UP

Carbon treatment and pH adjustment mean handling **hydrogen peroxide, acid, and hot nickel solution** — and dummy plating means the rectifier is live. Full PPE, ventilation, and LOTO for any electrical work. This is skilled maintenance work; do it under supervision until you're signed off.

CHAPTER 7

PPE: YOUR DAILY ARMOR

THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT HARMS, SPLASHES THAT BLIND

A nickel line works with hot caustic, strong mineral acids, electrical current at high amperage, chemical mists, and nickel chemistry that permanently sensitizes skin. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

**REMEMBER**

There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes — and your skin's tolerance for nickel — cannot.

• THE MASTER PPE SET (WEAR AT ALL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — *over* the goggles for any work involving acids, hot caustic, or the plating bath.
- **Chemical-resistant gloves** — elbow-length, rated for acids and alkalis (nitrile, neoprene, or PVC). Inspect for pinholes before each use. **On a nickel line, gloves are non-negotiable at every station** — even dilute nickel sensitizes skin.
- **Chemical-resistant apron** — full-length, chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof. Floors near plating tanks get wet and slick.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes are likely.

**HEADS-UP – NICKEL IS A SKIN SENSITIZER**

Repeated skin contact with nickel — even weak, dilute solutions — can cause **allergic contact dermatitis, known as “nickel itch.”** Once you become sensitized, it is **cumulative and permanent** — you can react to nickel for the rest of your life, on the job and off. Wear gloves at *every* station where nickel is present, including the post-plate rinse where the solution is dilute. Do not let familiarity make you casual about bare-hand contact.

**HEADS-UP – NICKEL MIST AND DUST**

Air agitation in the plating tank throws up a fine nickel-containing mist. **OSHA PEL for soluble nickel compounds is 1 mg/m³ TWA (as Ni).** Nickel compounds are classified as suspected carcinogens. Local exhaust ventilation is mandatory over the tank; mist eliminators are recommended. When charging dry nickel salts or boric acid, avoid raising dust. Use the respirator (P100 or supplied-air) your shop specifies when ventilation is inadequate.

**HEADS-UP – BORIC ACID**

Handle boric acid as a **reproductive-hazard concern**. Avoid inhaling the dust when charging or making additions, wear gloves, and wash after handling.

**TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

• STATION-BY-STATION HAZARD SUMMARY

STAGE	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Soak Clean / Electroclean	Hot alkaline (caustic), 140–180°F; H ₂ /O ₂ gas from electroclean	Chemical burns; hot splashes; flammable gas
02 Rinse	Residual alkaline cleaner	Mild skin/eye irritant; wet/slip hazard
03 Acid Activation	Strong mineral acid; HCl fumes (5 ppm ceiling); flammable H ₂ ; HF if die-cast option	Severe burns & eye damage; corrosive fumes; fire risk; HF is extreme
04 Rinse	Dilute acid residue	Mild irritant; slip hazard
05 Semi-Bright Nickel Plating	Hot acidic nickel bath; nickel mist (1 mg/m ³); high-amperage rectifier	Burns; skin sensitization; suspected carcinogen mist; shock/arc-flash
06 Rinse	Dilute nickel residue	Skin sensitizer even when dilute; slip hazard
07 Post-Treatment	Bright Ni + chrome tanks (their own hazards); bake oven (burns/fire); Cr(VI) at the chrome stage	Burns; fire; Cr(VI) is a confirmed carcinogen
08 Final Rinse/Dry	Hot water/air; residual chemistry	Burns from hot dryer; slip hazard

**HEADS-UP**
Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting plating chemicals — especially nickel. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 8

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read. **Locate, before your first shift on the line:** the nearest **emergency eyewash** (within ~10 seconds' reach of every chemical station), the nearest **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect** and main power shutoff, and the **spill kit** and **SDS binder** (Safety Data Sheets — the official hazard documents for every chemical on site).

**HEADS-UP**
Eyewash and shower stations must always be unblocked. Never stack parts, totes, or drums in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Chemical on skin	Get to the safety shower; flush at least 15 minutes . Remove contaminated clothing while flushing — don't stop early because "it feels better." Get medical attention; bring the SDS.
Chemical in eyes	Go to the eyewash; hold eyelids open and flush at least 15 minutes , rolling your eyes. Do not rub. Do not stop early. Get medical attention immediately — eye exposures are always urgent. Bring the SDS.
Inhaled fumes/mist, feel ill	Move to fresh air immediately. Report it — do not "tough it out." Seek medical attention for anything beyond brief, mild irritation. Bring the SDS.
Chemical swallowed	Do not induce vomiting unless the SDS directs it (vomiting acid or caustic burns the throat again). Follow SDS first aid; call medical help / poison control immediately.
Electrical shock / arc-flash	Do not touch the person if they may still be in contact with current. Cut power at the emergency stop / disconnect first, then call for help and render aid.
Chemical spill	Alert others and keep people away. Only clean it if trained and within your shop's "small spill" limits — otherwise evacuate and call trained spill response. Never let spills reach floor drains.

**HEADS-UP – HYDROFLUORIC ACID (HF)**

If your shop uses the fluoride activator for zinc die cast — common on decorative lines — know that **HF is uniquely dangerous**: it penetrates skin and attacks bone, and can cause harm from exposures that don't hurt at first. A specific first-aid protocol and **calcium gluconate gel** must be on hand *before* HF is used. Any HF exposure is a medical emergency — flush and get help immediately.

**HEADS-UP – LOCKOUT/TAGOUT (LOTO)**

LOTO is mandatory for *all* rectifier and electrical maintenance. LOTO means physically locking the power off and tagging it so no one can switch it back on while someone is working. Never work on a rectifier that hasn't been locked out — "I'll just be a second" is how people get electrocuted.

**HEADS-UP – A CRITICAL ACID-MIXING RULE, FOR LIFE**

When diluting acid, **always add ACID to WATER — never water to acid**. Adding water to concentrated acid releases heat so violently it can boil and erupt the acid into your face. The memory aid: "**Do as you oughta — add acid to water.**" Pour slowly, down the side, with the acid going *into* a larger volume of water.

**FUN FACT**

The reason acid-to-water matters is heat capacity: a big pool of water absorbs the heat of dilution when the acid is the small addition. Get it backwards and the tiny bit of water flash-boils and spits acid. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 9

THE SAFETY PHILOSOPHY

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Five principles guide everything on this line.

1. ASSUME EVERY TANK CAN HURT YOU.

Even the rinse tanks carry residual chemistry — and on this line even a *dilute* nickel rinse can sensitize your skin. Even “just water” can be slippery enough to put you on the floor. Respect comes first; familiarity is where complacency — and accidents — begin.

2. CONTAINMENT BEATS CLEANUP.

Move parts deliberately. Let them drain over the tank before transferring. Don't fling racks, splash, or rush handoffs. Most chemical exposures aren't dramatic spills — they're splashes and drips from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out means cleaner rinses, a longer-lived bath, and a firewall that actually holds).

3. RESPECT THE TWO FACES OF EVERY HAZARD.

On this line, the things that make good parts are the same things that can hurt you. The acid that activates the metal burns your skin. The current that deposits nickel can arc-flash. The nickel that protects the part sensitizes your skin and its mist is a suspected carcinogen. You can't separate the work from the hazard — so you manage both at once, every time.

4. ON A DUPLEX LINE, DISCIPLINE PROTECTS TWO THINGS AT ONCE.

Because the nickel bath runs for years, the same care that keeps it healthy — careful rinsing, no shortcuts, controlling drag-in, wearing gloves — is the care that keeps *you* healthy. And here that discipline carries a third passenger: the firewall between the two nickel chemistries. The habits are one and the same. An operator who rushes handoffs splashes chemistry on themselves, poisons rinses, *and* punches holes in the firewall — one sloppy motion, three kinds of damage.

5. WHEN IN DOUBT, STOP AND ASK.

No one expects you to have every answer, especially while learning. The unsafe operator is not the one who asks questions — it's the one who guesses. If a fume seems strong, a glove looks worn, a reading looks wrong, an unlabeled bottle appears at your tank, or a procedure feels off, *stop and check*.

• THE THREE HAZARD FAMILIES TO INTERNALIZE

- 1. **Acids & Alkali (corrosives).** The hot caustic cleaner (Stage 1) and the mineral acids (Stage 3) cause burns on contact and damage eyes instantly. The nickel bath itself is hot and acidic. *Defense:* goggles, face shield, gloves, apron — the 15-minute flush rule if exposed — and never forget acid-to-water.
- 2. **Electrical & physical.** The rectifier runs high amperage — shock and arc-flash risk. Wet floors mean slips. Bake ovens burn. *Defense:* LOTO for all electrical work, never touch a rectifier or bus bar with wet hands, non-slip boots, keep walkways clear and dry.
- 3. **Nickel chemistry (sensitizer / suspected carcinogen).** Nickel sensitizes skin permanently (“nickel itch”) and its mist is a suspected carcinogen. *Defense:* gloves at every nickel station, local exhaust over the tank, respirator when ventilation is inadequate, and strict no-hand-to-mouth hygiene. *(And if your duties ever extend to the chrome stage downstream, Cr(VI) has its own, stricter rulebook — see the Decorative Chrome series.)*

★

REMEMBER

Acids and alkali; electrical and physical; nickel chemistry. Three families. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

★

REMEMBER — THE WHOLE FOUNDATION

You’ve now got the whole foundation: what plating is, what makes semi-bright nickel special (and how it differs from both of its siblings), why the duplex stack outlasts a single layer and how the STEP test proves it, how the eight-stage line works as one connected system, why the order is locked by chemistry, the bath and its leveler in depth, the firewall discipline, what to wear, what to do in an emergency, and the safety mindset that ties it all together. Turn the page knowing the *why* — and the *how* will make a lot more sense.

ALKALINE SOAK CLEAN / ELECTROCLEAN

STRIP EVERY TRACE OF OIL, OXIDE, AND BUFFING COMPOUND – THE WHOLE DUPLEX STACK IS ONLY AS SOUND AS THE SURFACE UNDER THE FIRST NICKEL LAYER

Welcome to the first working tank. Semi-bright nickel is usually applied over **polished or buffed decorative substrates** — parts arriving loaded with buffing compound, drawing oils, and oxide. Buffing-compound residue is one of the most common causes of duplex-line pitting and skip plate, which makes cleaning frequently the **most demanding step in the whole line**. And remember the leveler’s limit: **semi-bright levels scratches, not dirt**. A poorly cleaned surface plates as pits, skip plating, or peeling under the base layer — and every fault carries up through the bright nickel and chrome.

WHAT YOU’RE DOING & WHY IT MATTERS

Cleaning happens in two moves:

- **Soak clean** removes the *bulk* soil — oils, grease, buffing compound, shop grime — by soaking in a hot alkaline (caustic) cleaner.
- **Electroclean** scrubs off the *microscopic* film that soak-cleaning leaves behind, by running electrical current through the part in the cleaner. The gas bubbles it generates physically lift the last of the soil off the surface.

The whole point is a **water-break-free** surface: one where water sheets off in a continuous, unbroken film. If water beads up, oil remains.

★
REMEMBER – THE WATER-BREAK TEST IS FREE AND NON-NEGOTIABLE
Pull a cleaned part and watch how water sheets off it. A smooth, continuous, unbroken sheet (“zero seconds” to break) means the surface is clean. If water beads or “breaks” into droplets, there’s still oil there — **stop and re-clean**. It’s the simplest, most powerful quality check in the whole shop.

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Soak Clean Temp	140–180°F (60–82°C)	Do not exceed 200°F — damages some cleaners
Soak Clean Time	3–10 min	Heavy draw oil: 8–10 min. Light stamping oil: 3–5 min
Soak Concentration	4–8 oz/gal (30–60 g/L)	Titrate weekly minimum
Electroclean Polarity	Cathodic → finish Anodic (last 15–30 s)	Anodic last strips smut back off
Electroclean Voltage	4–8 V	Current density 30–80 ASF typical
Electroclean Time	1–3 min	High-strength steel (>40 HRC): prefer anodic-only
Electroclean Temp	140–180°F	Same cleaner or dedicated electrocleaner
pH	11.0–13.5	Depends on formulation
Water Break Test	Pass = continuous sheet	Mandatory before proceeding

**TECHNICAL STUFF – CATHODIC VS. ANODIC CLEANING**

"Cathodic" means the part is negative (like during plating); it generates about *twice* the gas volume — more scrubbing — but can deposit metallic smut onto the surface. "Anodic" means the part is positive; less gas, but it strips that smut back off. That's why the standard move is **cathodic first, then finish anodic for the last 15–30 seconds** — you get the scrubbing power, then clean up the smut.

**HEADS-UP – HIGH-STRENGTH STEEL: ANODIC-ONLY**

Cathodic cleaning loads hydrogen *into* the steel. On high-strength steel (>40 HRC), prefer **anodic-only** electrocleaning — cathodic loads hydrogen, and the bake is still required downstream. (This just keeps you from adding more hydrogen than you have to.)

**HEADS-UP – ZINC DIE CAST: GO GENTLE**

On zinc die castings — very common in decorative work — limit cathodic time to prevent hydrogen blistering of the casting, and use a mild cleaner at lower temperature. An aggressive hot caustic can etch and discolor die cast. (This die-cast rule is about blistering the casting, not H-embrittlement — a different rule for a different reason than the >40 HRC one above.)

• STEP-BY-STEP PROCEDURE

1. **Confirm ventilation** and put on full PPE (goggles, face shield, elbow-length gloves, apron, boots).
2. **Verify temperature (140–180°F) and concentration** (titrate; 4–8 oz/gal). Skim any floating oil off the surface first.
3. **Immerse the rack fully** and soak for the part's soil load (3–10 min; heavily buffed parts get the long end).
4. **Electroclean:** run cathodic for the bulk of the time, then **finish anodic for the last 15–30 seconds**. For >40 HRC steel, run anodic-only.
5. **Pull and run the water-break test.** Continuous sheet = pass. Beading = re-clean; do not proceed.
6. **Transfer promptly** to the pre-activation rinse — a clean part left in air starts to re-film.

**HEADS-UP**

The cleaner is **hot (140–180°F) and strongly alkaline (pH 11–13.5)** — it causes chemical burns and can seriously injure your eyes. Electrocleaning evolves flammable hydrogen and oxygen gas at the surface. Full PPE, ventilation running, and know where your shower and eyewash are before you need them.

• TEACH-BACK CHECKLIST

- ☐ Explain what soak clean removes vs. what electroclean removes (bulk soil vs. microscopic film).
- ☐ Perform and interpret the water-break test (continuous sheet = clean; beading = oil remains).
- ☐ Explain why electroclean finishes anodic (to strip metallic smut back off).
- ☐ State when to use anodic-only (high-strength steel >40 HRC, to limit hydrogen loading) — and why the die-cast “limit cathodic” rule is a different rule.
- ☐ Explain why buffing compound is this line’s special enemy, and why the leveler can’t save a dirty part (“levels scratches, not dirt”).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp	Titrate cleaner; increase temp
Oily film after cleaning	Floating oil redepositing; exhausted cleaner	Skim/dump; recharge cleaner
Buffing-compound haze remains	Heavy polishing residue not emulsified	Extend soak; use a compound-removing cleaner; add electroclean
White residue on parts	Hard water salts; heavy drag-out	Check water quality; improve rinse
Etching of zinc die cast	Cleaner too alkaline/hot for substrate	Reduce temp; use mild cleaner
Pitting in nickel downstream	Organic residue not removed	Add/extend electroclean; verify water break
Skip plating (bare spots)	Passive oxide film; fingerprints	Electroclean with anodic finish; handle with clean gloves

SIGN-OFF — STATION 01

Trainee: _____ Trainer: _____ Water-break: PASS / FAIL • Anodic-only used? Y / N

“I confirm I ran the soak clean at temperature and concentration, electrocleaned with an anodic finish (or anodic-only for high-strength steel), and verified a water-break-free surface before transferring.”

STATION 02 OF 08

RINSE (PRE-ACTIVATION): THE FIREWALL

STRIP THE ALKALINE CLEANER FILM BEFORE THE ACID DIP – OR NEUTRALIZE IT IN THE WRONG TANK

You just cleaned the part in a hot, alkaline (basic) cleaner. The next tank is acidic. If you carry cleaner straight into the acid, it neutralizes and wastes the acid, and drags surfactants, silicates, and sodium downstream toward the nickel bath. This rinse is the **firewall** between the cleaning chemistry and everything after it. (It won't be the last firewall you meet on this line — Station 06 guards an even touchier border.)

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F)	No heating required
Time	30–60 s with agitation	Longer for blind holes, tubing, recesses
Water Source	Municipal or DI	DI preferred for nickel lines
Conductivity	< 500 µS/cm (target < 300)	Monitor daily
Flow Rate	Continuous overflow or counter-current	2–4 gal/min typical
pH	7–10	Elevated pH = cleaner drag-in

**REMEMBER – THE “CONTAMINATION NUMBER”**

Conductivity, in µS/cm, tells you how much dragged-in chemistry is in the rinse. Higher = dirtier. Here the target is **< 500 µS/cm**. If it creeps up, you're dragging cleaner through — increase flow or add a rinse stage.

**HEADS-UP**

The water carries residual alkaline cleaner. Wear goggles, chemical gloves, and non-slip footwear. Wet floors are a slip hazard.

• TEACH-BACK CHECKLIST

- ☐ Explain why an alkaline cleaner and an acid must never meet on the part (neutralization wastes acid; drag-in pollutes downstream).
- ☐ State the conductivity target (< 500 µS/cm) and what a rising number means.
- ☐ Explain why blind holes and recesses need extra rinse time.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
High conductivity	Excessive drag-out; low flow	Increase drain time over cleaner; increase fresh water flow
Soapy residue on parts	Inadequate rinse time/volume	Extend dwell; add agitation; consider double rinse
Acid dip goes alkaline quickly	Cleaner carry-over overwhelming acid	Tighten rinse; add second rinse stage
Foaming in rinse	Surfactant from cleaner	Increase water flow; check cleaner formulation

SIGN-OFF – STATION 02

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

“I confirm I rinsed to below the conductivity target with agitation, giving recesses extra time, before moving to the acid dip.”

STATION 03 OF 08

ACID ACTIVATION (THE ACID DIP)

DISSOLVE THE OXIDE LAYER AND EXPOSE BARE METAL FOR NICKEL ADHESION

Even a perfectly clean surface grows an invisible **oxide** film in air. Nickel can't bond to oxide — only to bare, chemically active metal. The acid dip dissolves that oxide. **Under-dip** and oxide remains, so the nickel peels. **Over-dip** and you etch the part and load it with hydrogen. In a duplex system, adhesion of the semi-bright base layer carries the *entire stack* — oxide is invisible, but adhesion failure at the base of a three-layer stack is not, and it fails the whole part.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient–110°F	Ambient typical for mild activation
Time — Steel	15–60 s	Light oxide 15–30 s; heavy scale 30–60 s
Time — Brass/Copper	5–15 s	Short dip — acid attacks copper alloys fast
Time — Zinc Die Cast	5–15 s	Minimal; base metal dissolves rapidly; fluoride activation preferred
Acid (Option A)	HCl 10–50% v/v	Common; good oxide removal; fumes (5 ppm ceiling)
Acid (Option B)	H ₂ SO ₄ 5–20% v/v	Less fuming; slower; no chloride introduced
Acid (Option C)	10–20% H ₂ SO ₄ + 1–2% HF (die cast)	Activates die cast without excessive base-metal attack
Iron in bath	Dump when > 50 g/L Fe	Spent acid loses effectiveness; drags metals into nickel
Agitation	Manual rack movement	No air agitation — acid mist hazard

• SUBSTRATE-SPECIFIC NOTES

- **Steel / cast iron:** Standard HCl or H₂SO₄ activation. Use **inhibited acid** for >40 HRC steel to reduce hydrogen absorption.
- **Brass / copper alloys:** Short dip only (5–15 s). Excessive time dissolves copper into the acid, which drags into the nickel bath. Decorative brass often takes a copper strike ahead of the semi-bright layer — follow the system spec.
- **Zinc die cast:** Extremely sensitive — and very common in decorative work. Use a dilute fluoride-containing activator per supplier TDS — do **not** use strong HCl. After activation, die cast takes a **copper strike** (cyanide or alkaline non-cyanide) *before* the semi-bright nickel — never directly into the acid nickel bath, or you get an immersion deposit, peeling, and zinc contamination.

• THE BIG ONE: HYDROGEN EMBRITTLEMENT



HEADS-UP – SAFETY & QUALITY: HYDROGEN EMBRITTLEMENT

When acid attacks steel, it releases hydrogen. Most bubbles away as gas — but some sneaks *into* the steel as single atoms, wedging into the crystal lattice. On ordinary mild steel this is usually harmless. But on **high-strength / hardened steel (≥ 40 HRC / ≥ 180 ksi UTS)**, trapped hydrogen makes the metal **brittle** — and a brittle part can **crack and fail suddenly under load**, sometimes hours or days later.

The acid dip is the single greatest source of hydrogen absorption on the entire line. Uninhibited acid on hard steel can load the lattice *in seconds*.

What you do about it: use the **minimum effective acid concentration and dip time** — don't over-dip "to be safe" (that's the opposite of safe here). **Always use an inhibitor on substrates over 40 HRC.** If a job is flagged high-strength, **stop and check with your lead** — and know that hardened parts require a post-plate **bake** (Station 06/07) per ASTM B850.

Terms: **HRC** = Rockwell C hardness (higher = harder). **ksi UTS** = thousands of psi of ultimate tensile strength. You don't calculate these — just treat them on a job traveler as a "slow down and check" flag.

• STEP-BY-STEP PROCEDURE

1. **Check the job for hardness flags first.** High-strength / hardened / ≥ 40 HRC / ≥ 180 ksi? Confirm the right process and that inhibitor is in use before anything else.
2. **Confirm ventilation is running** and your PPE (including face shield) is on.
3. **Verify acid strength and dip time** for this substrate (15–30 s clean steel; 30–60 s heavy oxide; 5–15 s copper alloys / die cast).
4. **Immerse fully and start your timer.** Agitate by gently moving the rack by hand — **never air**.
5. **Watch the surface.** Light bubbling is normal; the dark oxide should visibly lighten. If it isn't clearing, the acid may be dilute or spent — don't just leave a hard part in longer; check the bath.
6. **Pull at the timed mark.** Don't "round up." Over-dipping etches and embrittles.
7. **Drain briefly** and move **promptly** to the pre-plate rinse — activated metal re-oxidizes fast in air.

• TEACH-BACK CHECKLIST

- ☐ Explain what the acid dip removes and why nickel can't bond without it (oxide; no activation, no adhesion).
- ☐ Explain over-dip vs. under-dip consequences (etching + embrittlement vs. oxide remains + peeling) — and why base-layer adhesion carries the whole stack.
- ☐ State dip times for clean steel vs. heavy oxide vs. copper alloys / die cast.
- ☐ Explain why there is NO air agitation here (acid mist hazard).
- ☐ Explain hydrogen embrittlement and which parts are at risk (≥ 40 HRC / ≥ 180 ksi), and the two things that reduce it (minimum dip, inhibitor).
- ☐ Recite the acid-dilution rule (always add acid to water).

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Dark oxide remains	Acid too dilute; time too short	Increase concentration/time (<i>check embrittlement risk first</i>)
Pitting / etching	Too strong, too long, or too warm	Reduce concentration; shorten dip; add inhibitor
Adhesion failure / blistering of the stack	Incomplete activation; residual oxide	Re-evaluate acid strength/time; check for alkaline drag-in
Copper smut on brass	Dezincification — acid attacks zinc in alloy	Shorten dip; use copper strike before nickel
White smut on die cast	Fluoride too high or too long	Reduce concentration; shorten time; follow TDS
Excessive HCl fuming	High concentration/temp; poor ventilation	Reduce to minimum effective; check exhaust



HEADS-UP

Strong mineral acid — severe burns and eye damage. HCl fumes at room temperature (5 ppm ceiling). Hydrogen gas from pickling is flammable. If HF is used for die cast, it is extremely hazardous — calcium gluconate gel must be on hand. Full acid PPE and ventilation.

SIGN-OFF — STATION 03

Trainee: _____ Trainer: _____ Dip used: _____ s • Inhibitor? Y / N

"I confirm I checked the job for hardness flags, ran the acid dip at the timed interval with manual agitation only, ventilation and PPE in place, and used inhibitor where required."

STATION 04 OF 08

RINSE (PRE-PLATE): THE GUARD RINSE

THE LAST BARRIER BEFORE YOUR SEMI-BRIGHT BATH — A TANK THAT MAY RUN FOR YEARS WITHOUT A DUMP

You're at the last station before the semi-bright plating bath — the crown jewel of the line. Your job here is to **wash the acid (and the dissolved iron and metals it carries) off the part** so it doesn't contaminate that expensive, maintained-for-years bath. Think of this rinse as the **guard at the door** of the most valuable room in the building.

WHAT YOU'RE DOING & WHY IT MATTERS

Coming out of the acid dip, the part is wet with acid — and that acid now carries **dissolved iron** (and possibly copper or zinc) it picked up while pickling. Carry that into the semi-bright bath and you cause three problems:

- 1. **You drop the bath's pH.** The bath runs in a tight window (3.5–4.5); extra acid pushes pH down, changing stress and risking burning.
- 2. **You add iron.** Iron accumulates, causing haze, roughness, raised stress, and reduced ductility — directly attacking the properties that make semi-bright a sound duplex base — and it's hard to remove.
- 3. **You add other metals** (Cu, Zn) that accumulate the same way.

★

REMEMBER — EVERY CONTAMINANT HERE IS A PERMANENT GUEST
Because the semi-bright bath is maintained rather than replaced, every milligram of acid, iron, copper, or zinc that enters through poor rinsing accumulates — in the base layer of *every part you plate from then on*. This is the single most important contamination barrier ahead of the tank. Hold it to a tighter standard than any rinse before it.

OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient (60–85°F)	No heating required
Time	30–60 s with agitation	Double rinse strongly recommended
Water Source	DI strongly preferred	Municipal water adds calcium, chloride, more
Conductivity	< 200 µS/cm (target < 100)	Critical for nickel bath life
pH	4.0–8.0	Below 4 = excessive acid drag-in
Flow Rate	Continuous overflow, 3–5 gal/min	Counter-current ideal
Drain Time	15–20 s minimum	Prevents diluting the nickel bath

**REMEMBER — THE 30-SECOND CLOCK**

Move into the nickel bath **within 30 seconds** of leaving the rinse. A freshly activated, rinsed surface starts to oxidize in air almost immediately, and an oxidized surface plates poorly — undoing Station 3's work. Keep the part wet; speed matters at this handoff.

**HEADS-UP**

The water carries dragged-in acid — diluted but still irritating. Goggles, chemical gloves, non-slip footwear. Parts must move quickly to the nickel tank — no delays.

• TEACH-BACK CHECKLIST

- ☐ Explain the three ways acid/iron drag-in hurts the semi-bright bath (pH drop, iron, other metals).
- ☐ State the conductivity target (< 200 $\mu\text{S}/\text{cm}$) and why it's tighter than Station 02.
- ☐ Explain the 30-second rule and why re-oxidation matters.
- ☐ Explain why double-rinsing is recommended here specifically.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Conductivity high	Heavy acid drag-out; low flow	Increase drain time from acid dip; increase flow
Nickel bath pH dropping	Acid drag-in from activation	Tighten conductivity target; add second rinse
Iron climbing in nickel bath	Dissolved Fe from spent pickle	Refresh pickle; improve rinse; add a stage
Parts oxidizing before plating	Transfer too slow; parts drying	Speed transfer — enter nickel within 30 s
Chloride climbing in nickel bath	HCl drag-in	Switch to H_2SO_4 activation; or improve rinse

SIGN-OFF — STATION 04

Trainee: _____ Trainer: _____ Conductivity: _____ $\mu\text{S}/\text{cm}$ • To-tank time under 30 s? Y / N

"I confirm I rinsed below the tight conductivity target (double rinse where available), drained well, and moved the part to the nickel bath within 30 seconds."

**REMEMBER — THE PREP-LINE PROMISE**

Four stations, one mission: hand the semi-bright bath a surface that is **clean, contaminant-free, oxide-free, and still wet**. Clean it (01), rinse the cleaner off (02), strip the oxide (03), rinse the acid and iron off (04), and move fast. Every one a checkpoint, not a rest stop. Do these four right, every rack — and a sound base layer will reward you. Cut corners here, and no amount of bath chemistry downstream can save the stack.

PART THREE — THE MAIN LINE (STATIONS 05–08)

STATION 05 OF 08

SEMI-BRIGHT NICKEL PLATING: THE MAIN TANK

A WATTS-TYPE BATH WITH A SULFUR-FREE LEVELER — AND NOTHING ELSE

This is the payoff. With a perfectly clean, freshly activated, well-rinsed part, the base layer of the duplex stack goes on. The bath: nickel sulfate, nickel chloride, boric acid, the sulfur-free leveler, an anti-pit wetter — leveled, ductile, sulfur-free, and more noble than the bright layer above it, *by design*. You already met this bath in depth in Chapter 6 — this chapter is how you *run* it, shift to shift.



REMEMBER — WHAT YOU'RE BUILDING

A **leveled, satin, essentially sulfur-free (<0.005% S)** deposit that is deliberately **more noble** than the bright nickel that will be plated over it. If it comes out satin with the polish lines filled, that's *success*. You are not chasing shine — the mirror is the bright tank's job. *"No sulfur, no mirror — on purpose."*

• OPERATOR ESSENTIALS (SEE CH 6 FOR THE FULL WINDOW TABLE)

- **Temperature:** 120–150°F (target 130–140°F — warmer = lower stress, higher ductility; *steady* = consistent leveling).
- **pH:** 3.5–4.5. Lower with H_2SO_4 , raise with nickel carbonate — **never NaOH**.
- **Current density:** 20–50 ASF rack; 5–15 ASF barrel. Stay in the window — too high burns edges and raises stress.
- **Boric acid:** keep above 30 g/L (it buffers against burning).
- **Sulfur-free leveler:** dosed on **amp-hours** per TDS; confirmed by leveling panel — never by eye, never by calendar.
- **Air agitation + filtration:** both running continuously — steady agitation keeps leveling uniform and lowers pitting; filtration keeps particulate out.
- **Anodes:** per the sulfur-content spec (Ch 6.5) — S-Rounds for general work, non-sulfur where the layer's sulfur content is tightly specified (then keep chloride adequate and anode area generous). Doubled anode bags always.

• THE RULE CARD: < 0.005% SULFUR IN THE DEPOSIT



TECHNICAL STUFF

Keeping sulfur out of the deposit (<0.005% S) is what makes the semi-bright layer more noble than the bright nickel plated over it. That nobility difference — verified by the ASTM B764 STEP test — is why the bright layer corrodes sacrificially and the duplex stack outlasts a single nickel layer of the same thickness. Add sulfur — wrong additive, bright-bath drag-over, or anode residues where a sulfur-free system is specified — and you lose the whole mechanism. (Cathode efficiency here is 93–97%, same as the family — but on this tank, the number that defines the job is the sulfur ceiling, not the efficiency.)

• STEP-BY-STEP PROCEDURE

1. **Confirm PPE and ventilation.** Gloves at all times (nickel sensitizer); local exhaust on (nickel mist).
2. **Check temperature, pH, and boric acid** at shift start; confirm agitation and filtration are running; check the amp-hour log against the leveler-addition schedule.
3. **Confirm the CD / plating time** for the target thickness on the job ticket. Enter the tank within 30 s of the pre-plate rinse.
4. **Immerse, connect to the cathode bar, and start.** Watch for even gassing and no arcing at contacts.
5. **Plate to the target thickness** per the job spec — on duplex work the semi-bright thickness and its share of the total nickel are set by ASTM B456 / the customer spec, not by eye.
6. **Log the amp-hours.** The leveler is consumed by the work; the log is how the next addition lands on time.
7. **Pull, drain over the tank** (recover drag-out), and move to the post-plate rinse — the firewall crossing.
8. **Log the run.** Note anything off — poor leveling, gray patches, burning, pitting, roughness — for the next panel / maintenance check.

• READING TROUBLE (SEE CH 6.7 FOR THE FULL PANEL GUIDE)

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Burning at high CD	pH too HIGH; boric acid or Ni metal low; temp too LOW; CD too high	Check pH, boric acid, Ni metal, temp; lower CD
Poor leveling	Leveler low/depleted; too much drag-out	Add sulfur-free leveler per Ah; tighten drag-out; re-panel
Dull gray areas the bright layer won't cover	Coumarin breakdown products; organic (oil) drag-in	Carbon treat on an Ah schedule; check for oil drag-in
Deposit dulls / bright layer patchy after plating	Sulfur pickup in semi-bright (wrong additive, bright-bath drag-over, or anode residues where sulfur-free is specified)	Stop cross-contamination; verify additive; review anode choice; STEP-test to confirm layer potentials
Pitting (gas pits)	Low anti-pit wetter; poor agitation; organics	Add wetter; increase air; carbon treat; check filter
Roughness / nodules	Particulate; poor filtration; anode fines	Check filters; replace anode bags; raise filtration
Peeling / poor adhesion	Poor cleaning; incomplete activation; oxidized surface	Recheck water break; recheck acid dip; speed transfer
Raised stress / cracking	Iron/metallic contamination; CD too high; chloride too high	Analyze metals; carbon treat; lower CD; check chloride; raise temp
Poor throwing power / uneven thickness	Low NiCl ₂ ; pH too low; low temp; poor agitation	Adjust chloride within the semi-bright target; optimize pH/temp; improve agitation
Anodes passivating (voltage climbing)	Low chloride; too little anode area; non-sulfur anodes at high CD	Raise chloride within target; add anode area; review anode type against the sulfur spec
Duplex fails corrosion test despite good thickness	STEP potential difference collapsed — sulfur pickup, or layer thicknesses/ratio wrong	STEP-test both layers; eliminate sulfur cross-contamination; verify thickness split per ASTM B456 / customer spec

**HEADS-UP**

Hot acidic nickel solution (pH 3.5–4.5, up to 150°F) — causes burns and is a **skin sensitizer**; nickel compounds are **suspected carcinogens**. “Nickel itch” is cumulative and permanent once sensitized — minimize skin contact always. Air agitation generates **nickel mist** — **local exhaust mandatory (PEL 1 mg/m³ as Ni)**. Rectifier = high amperage — **LOTO for all maintenance**. Handle **boric acid** per its reproductive-hazard note (no dust, gloves, wash after). PPE: goggles + face shield, elbow-length nitrile gloves, apron, boots, P100 or supplied-air respirator if ventilation is inadequate.

**HEADS-UP — TANK MATERIAL**

This is a hot, acidic, chloride-containing bath. Tanks, heaters, and fixtures must be **PP, CPVC, or lined** — **never bare mild steel**, which the bath would corrode (dumping iron into your bath and failing structurally). Plain PVC is marginal above 140°F, so CPVC is the safer plastic for this hot bath.

• TEACH-BACK CHECKLIST

- ☐ Name the four levers plus the additive (metal, pH, temperature, CD + the sulfur-free leveler) and what each does.
- ☐ Explain why satin is correct here, where the mirror actually comes from, and why you never dose bright-bath additive into this tank.
- ☐ Explain amp-hour dosing — why the leveler is tracked on Ah, not the calendar — and how you confirm it (leveling panel, not brightness).
- ☐ State the pH range and the correct way to raise/lower it (nickel carbonate up, H₂SO₄ down; never NaOH).
- ☐ State the burning rule from memory: pH too HIGH, temp too LOW (plus low boric / low metal / high CD).
- ☐ Explain why the deposit is ductile but the steel still needs the bake.
- ☐ State the nickel PEL (1 mg/m³ as Ni) and the sensitizer / suspected-carcinogen hazards.
- ☐ Explain why the tank must be PP/CPVC/lined, not mild steel.

SIGN-OFF — STATION 05

Trainee: _____ Trainer: _____ Temp: _____ °F · pH: _____ · CD: _____ ASF · Ah
logged? Y / N

“I confirm I ran the bath in its temperature/pH/CD window with agitation and filtration, plated to the specified thickness, logged amp-hours for the leveler schedule, wore gloves with exhaust running, and made no additions except the correct semi-bright products per TDS.”

**TIP**

Keep a tank log and write in it every shift — temperature, pH, boric, amp-hours, additions, anything odd. A bath that runs for years builds its history in that log, and the day something drifts, the log is what tells you *when* it started and *what changed*. The best troubleshooting tool on this line costs a dollar and lives on a clipboard.

STATION 06 OF 08

POST-PLATE RINSE: THE TWO-CHEMISTRY FIREWALL

WASH OFF THE DRAG-OUT BEFORE THE NEXT LAYER — AND KEEP THE TWO NICKEL CHEMISTRIES FROM EVER MIXING

Fresh out of the semi-bright tank, the part carries **nickel drag-out** — nickel sulfate, chloride, and your sulfur-free leveler clinging to the surface. On a duplex line the part usually goes straight from here into the **bright** nickel bath, which makes this rinse the most consequential handoff on the line. It does triple duty: recover the nickel (a drag-out recovery rinse can capture it for return to the bath), **firewall the two chemistries** (your leveler must not reach the bright bath; their sulfur-bearing additive must not wash back toward yours), and serve as the **decision point for the hydrogen-embrittlement bake** on hardened steel.



REMEMBER

"On a duplex line this rinse does double duty: recover the nickel, and stop the two nickel chemistries from ever mixing." Chapter 6.4 gave you the firewall rules; this station is where you live them.

• OPERATOR ESSENTIALS

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating required
Time	30–60 s with agitation	Double rinse strongly recommended between the two nickel baths
Conductivity	< 200 µS/cm (target < 100)	Nickel drag-out AND leveler carryover are the concerns
Drain Time	15–20 s over semi-bright tank	Maximizes drag-back recovery; minimizes carryover forward
Drag-Out Recovery	Optional still first rinse	Periodically return to the semi-bright bath to recover Ni

• THE FIREWALL IN PRACTICE

- **Dedicated rinses only.** This rinse belongs to the semi-bright→bright crossing. Never route parts from other stages through it, and never let the bright stage share it.
- **Full drain, both directions.** Drain over the semi-bright tank before entering the rinse (recovers your nickel, cuts carryover forward); let the bright line worry about its own drain discipline coming back.
- **Watch the bright bath’s behavior.** If the bright deposits go dull or patchy and their team is chasing brighteners, ask when this rinse was last checked — leveler carryover through a tired rinse is the classic hidden cause.

• THE BAKE DECISION POINT



HEADS-UP – HYDROGEN-EMBRITTLEMENT BAKE

For hardened steel (> 40 HRC): **bake within 4 hours of the last electroplating step at 375 ±25°F for a minimum of 8 hours**, per ASTM B850. On a duplex/multilayer part the “last electrodeposition step” is usually the chrome or the bright nickel — **verify the customer specification** for whether the bake comes after the full stack or at an intermediate point. The 4-hour clock starts at completion of that last electrodeposition step. If baking is required at this stage, parts go from this rinse to the oven.



REMEMBER

The semi-bright *deposit* is ductile and low-stress — but the hydrogen absorbed by the *substrate* during cleaning, activation, and plating is the embrittlement concern, not the nickel. **A ductile deposit does not exempt a high-strength steel part from the bake.**

• TEACH-BACK CHECKLIST

- ☐ Explain the three jobs of this rinse (nickel recovery, chemistry firewall, bake decision).
- ☐ Explain both directions of the firewall failure (leveler forward = dull bright; sulfur back = collapsed STEP) and the rules that prevent them.
- ☐ State the bake rule for >40 HRC steel (375 ±25°F, ≥8 hr, within 4 hr, per ASTM B850) and where the clock starts on a multilayer part.
- ☐ Explain why a ductile deposit doesn't remove the bake requirement.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Bright bath dulling / leveler contamination	Semi-bright leveler carried forward	Tighten this rinse; add a dedicated recovery/dead rinse; increase drain time
Downstream tank life short	Nickel drag-in	Tighten this rinse; add recovery rinse
Water spots on nickel	Hard water; slow transfer	Use DI; minimize air exposure
Haze / staining on nickel	Oxidation during slow transfer; drag-out drying	Reduce transfer time; keep parts wet
Nickel salt crystallizing on rack	Inadequate rinsing; high drag-out	Extend rinse; strip racks more often

SIGN-OFF – STATION 06

Trainee: _____ Trainer: _____ Conductivity: _____ μS/cm · Bake required? Y / N

If Y, per spec at which step? _____

“I confirm I rinsed off nickel drag-out to target through the dedicated firewall rinse, and correctly identified whether and when the part required the H-embrittlement bake per the customer specification.”

STATION 07 OF 08

POST-TREATMENT: COMPLETING THE DUPLEX STACK

BRIGHT NICKEL AND CHROME OVER THE SEMI-BRIGHT BASE — THEN BAKE IF REQUIRED, AND STEP-VERIFY THE SYSTEM

Here is where semi-bright differs sharply from both siblings. Plain Watts parts often leave as-plated; bright nickel almost always ends in chrome. Semi-bright is the **foundation, not the finish** — for most work, “post-treatment” means the part goes on to the **bright nickel** tank and then to **decorative chromium**, completing the stack your layer anchors. Then the stack gets **verified** — because in a duplex system, looking right is not the same as being right.

• STACK-COMPLETION & POST-TREATMENT OPTIONS

TREATMENT	TYPICAL USE	KEY PARAMETER
Bright nickel over semi-bright (duplex)	Standard decorative duplex — the usual next step	Bright layer plated directly over semi-bright; total Ni thickness + split per ASTM B456 / customer spec
Microporous / microcracked Ni + Cr (triplex)	High-corrosion automotive / marine	Adds a thin high-activity nickel + microdiscontinuous chrome; per ASTM B456; STEP + porosity verification
Decorative chromium topcoat	Final appearance + tarnish/scratch resistance over the nickel stack	Decorative Cr 0.25–0.75 µm (10–30 µin); its own process — see the Decorative Chrome series
H-embrittlement relief bake	High-strength steel (>40 HRC)	375 ±25°F, ≥8 hr, within 4 hr of the last plating step (ASTM B850)
Clear lacquer / topcoat	Indoor decorative where chrome not used	5–15 µm; air or heat cure per product TDS
Chromate / conversion seal	Corrosion resistance without a chrome plate	Per supplier TDS; conversion coating, not electrodeposited

• RE-ACTIVATE BEFORE THE NEXT LAYER

Nickel that has sat around grows a thin passive film. If the part waits between the semi-bright tank and the bright tank — or between nickel and chrome — the surface **passivates**, and the next layer can peel. **Re-activate with a light acid dip after any delay**, and minimize the delay in the first place. On a smoothly-run duplex line the part moves semi-bright → rinse → bright with no waiting, and this never comes up. On a real line, with real interruptions, it comes up — and now you know the fix.



REMEMBER — RE-ACTIVATE BEFORE THE NEXT LAYER

Fresh layer over passivated nickel = peeling at the interface. Any delay between layers earns a light re-activation dip. Minimize the delay; when in doubt, re-activate.

• VERIFYING THE STACK

- **Thickness** — total nickel and the per-layer split, per ASTM B456 / customer spec.
- **STEP test (ASTM B764)** — where specified: confirms each layer's thickness *and* that the bright layer is anodic (less noble) to the semi-bright base, so it will corrode sacrificially. Right thicknesses, wrong potentials = a stack that fails corrosion testing. (Chapter 3.4 explains how it works.)
- **Adhesion** — qualitative methods per ASTM B571 (bend, thermal shock) where required.
- **Accelerated corrosion** — salt spray / CASS per the spec, on the schedule the customer requires.



HEADS-UP — HEXAVALENT CHROMIUM

If the finished stack calls for decorative chromium over the nickel, that chrome process carries its own severe hazards. **Hexavalent chromium (Cr⁶⁺) is a confirmed human carcinogen (IARC Group 1). OSHA PEL: 5 µg/m³ TWA (action level 2.5 µg/m³) as Cr(VI).** Full PPE, mist suppressant, local exhaust, and medical surveillance per OSHA 29 CFR 1910.1026 apply. This is the chrome line's responsibility — see the Decorative Chrome series — not the nickel line's. Know which chemistry runs where, and keep chrome drag-back OUT of both nickel baths (even 1 ppm Cr⁶⁺ wrecks a nickel deposit).

• FINAL RINSE & DRY (STAGE 08 ADDENDUM)

- DI water rinse, ambient, gentle immersion 15–30 s.
- Warm air dry: 120–140°F max.
- No cure time for the nickel itself; topcoats/lacquers cure per their own TDS.
- Handle with clean, dry, lint-free gloves — fingerprints on fresh nickel can stain or interfere with the next layer's adhesion.

• TEACH-BACK CHECKLIST

- ☐ Explain why semi-bright is “the foundation, not the finish,” and name the usual next two layers.
- ☐ Explain what the STEP test verifies and why right-thickness-wrong-potential still fails.
- ☐ Explain why nickel must be re-activated before the next layer after any delay.
- ☐ State the Cr(VI) hazard and PEL if a chrome topcoat is used, and whose responsibility it is.

• TOP MISTAKES & QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Bright layer won't cover / patchy over semi-bright	Semi-bright passivated on standing; not re-activated; sulfur pickup dulling the base	Re-activate with a light acid dip; minimize delay; STEP-test for sulfur pickup
Duplex fails salt-spray / CASS despite good total thickness	STEP potentials wrong; thickness split wrong; semi-bright picked up sulfur	STEP-test both layers; correct thickness split per B456; eliminate sulfur cross-contamination
Bright/chrome peels from semi-bright	Semi-bright surface oxidized; excessive transfer delay	Re-activate; reduce transfer time between layers
Dull gray showing through the bright layer	Coumarin breakdown / poor leveling in the semi-bright base	Fix at the semi-bright stage — carbon treat, restore leveler on Ah
Cracking after bake on thick stacks	Deposit already high-stress (contamination); bake relieves H, not deposit stress	Fix at plating — analyze metals, lower CD, check chloride
Stain/fingerprints between layers	Handling with bare or dirty hands	Clean lint-free gloves; keep parts clean, dry, re-activated



HEADS-UP

Bake ovens = burn and fire hazard — follow oven LOTO and hot-work procedures; no flammables near ovens. If bright nickel or a chromium topcoat is applied, their own hazards apply — nickel remains a skin sensitizer through all handling (gloves at all times), and Cr(VI) requires full controls per OSHA 1910.1026. Seal/lacquer chemistries: consult the product SDS.

SIGN-OFF — STATION 07/08

Trainee: _____ Trainer: _____ Next layer / treatment path: _____ · Re-activated after delay? Y / N / n-a · Dried without spots? Y / N

"I confirm I routed the part to the correct stack-completion or post-treatment path, re-activated where a delay preceded the next layer, handled with clean gloves, and dried the part cleanly."

PART FOUR — BACK MATTER & CERTIFICATION

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

IF A WORD ON THE FLOOR CONFUSES YOU, IT'S IN HERE

Activation (acid dip): A short dip in dilute acid that dissolves the invisible oxide film so nickel can bond to bare metal. Also used lightly ("re-activation") between layers after a delay.

Amp-hours (Ah): Current multiplied by time — the true measure of how much work the bath has done, and therefore how much leveler it has consumed. The leveler is dosed on Ah, never on the calendar.

Anode: The positive electrode — bars or baskets of nickel that dissolve to feed nickel into the bath.

Anode bag: A cloth bag around the anode that catches "fines" (bits of undissolved anode) so they don't roughen the deposit.

ASF (amps per square foot): Current density — how much current is spread over the area being plated. Metric version is A/dm².

ASTM B456: The standard for electrodeposited decorative **copper-plus-nickel-plus-chromium and nickel-plus-chromium** coatings — THE governing spec for the semi-bright + bright + chrome stack: service grades, layer thicknesses, thickness split, and verification (current revision). International equivalent: ISO 1456.

ASTM B571: The practice for qualitative adhesion testing — bend, peel, thermal-shock methods.

ASTM B764 (STEP test): The standard method for **S**imultaneous **T**hickness and **E**lectrode **P**otential determination of the individual nickel layers in a multilayer deposit — the measurement that proves the bright layer is anodic (sacrificial) to the semi-bright base.

ASTM B850: The standard for the post-plate **bake** that relieves hydrogen embrittlement in high-strength steel.

ASTM F519: Mechanical hydrogen-embrittlement evaluation (sustained-load test), where required.

Bake (H-relief): Heating a plated part (375 ±25°F, ≥8 hr, within 4 hr of the last plating step) to drive absorbed hydrogen out of high-strength steel.

Barrel plating: Plating many small parts at once in a rotating perforated barrel.

Boric acid: The pH buffer in the bath — holds acidity steady at the part's surface and prevents burning. Handle as a reproductive-hazard concern.

Bright nickel: The Watts-type bath run with a full brightener package (sulfur-bearing Class I carrier + Class II brightener) — mirror-bright, leveling, and deliberately *less noble* than semi-bright. The middle layer of the duplex stack. See the Bright Nickel manual.

Burning: A rough, dark, powdery deposit at high-current areas — caused by pH too HIGH, temperature too LOW, low boric acid, low metal, or too-high CD.

Carbon treatment: A periodic purification that removes organics — including leveler breakdown products — and iron from the bath using activated carbon and peroxide. On a semi-bright bath it runs on a regular amp-hour schedule.

CASS test: Copper-accelerated acetic-acid salt spray — an accelerated corrosion test used to qualify decorative nickel-chromium stacks per the spec.

Cathode: The negative electrode — your part, where nickel deposits.

Cathode efficiency: The fraction of current that deposits metal rather than making gas. This bath runs 93–97%.

• GLOSSARY (CONTINUED)

Conductivity ($\mu\text{S}/\text{cm}$): A measure of dissolved ions in rinse water — the “contamination number.” Higher = dirtier rinse.

Coumarin: The traditional sulfur-free leveler for semi-bright nickel. Consumed with the work and prone to breakdown — its degradation products cause dull gray, hard-to-brighten areas unless carbon-treated out.

Coumarin-free leveler: A modern proprietary sulfur-free leveler that approaches coumarin’s leveling with less troublesome breakdown. Dosing and carbon schedules per its own TDS.

Current density (CD): See ASF.

Drag-in / drag-out: The film of solution a part carries out of one tank (drag-out) and into the next (drag-in) — the reason rinses exist, and the vehicle for every firewall failure.

Ductility: How much a metal can bend or stretch before it cracks. The semi-bright base layer is highly ductile by design.

Dummy plating: Plating onto a scrap cathode at low current to pull metallic contaminants out of the bath.

Duplex nickel: A two-layer nickel system — more-noble semi-bright base under a less-noble bright layer — engineered so corrosion attacks the bright layer sideways instead of drilling down to the steel.

Electroclean: Cleaning with electrical current running through the part; gas bubbles scrub off the last microscopic film.

Firewall (duplex discipline): The set of rules — dedicated rinses, unshared dosing equipment, drag-out control, deliberate additions — that keeps the semi-bright and bright chemistries from ever mixing.

Hull Cell: A small angled test cell that plates a panel across a range of current densities at once — read here for *leveling and soundness*, not brightness.

Hydrogen embrittlement: Brittleness in high-strength steel caused by absorbed hydrogen; can cause sudden cracking under load. Relieved by baking.

Internal stress: Built-in tension (or compression) locked into a deposit as it grows; controlled by temperature, pH, CD, chloride, and contamination.

Ion: An atom carrying an electrical charge; nickel travels the bath as Ni^{2+} .

Leveler: An organic additive that fills scratches and polish lines by slowing deposition on peaks so valleys catch up. Semi-bright’s leveler is **sulfur-free** — that’s the whole point.

Leveling panel: A scribed or pre-scratched test panel plated at known CD to confirm the leveler is filling scratch lines — the semi-bright quality check that replaces “is it bright?”

LOTO (Lockout/Tagout): Physically locking and tagging power off before electrical maintenance.

Microporous / microcracked chrome: Chrome plated with thousands of engineered micro-openings that spread corrosion current across many tiny sites — part of triplex systems.

Nickel itch: Allergic contact dermatitis from nickel exposure — cumulative and permanent once sensitized.

Noble (more/less): How reluctant a metal surface is to corrode, expressed as electrode potential. The semi-bright layer is *more* noble (protected); the bright layer is *less* noble (sacrificial). Sulfur content is what sets the difference.

• GLOSSARY (CONTINUED)

Passivation (of anodes): When a non-sulfur anode goes inert and stops dissolving; prevented by adequate chloride and anode area. (Passivation *of the nickel surface* between layers is the reason for re-activation.)

Rack plating: Plating parts individually hung on a current-carrying frame.

Rectifier: The power supply that converts AC to the steady DC used for plating.

Sacrificial corrosion: When a less-noble metal in electrical contact with a more-noble one corrodes preferentially, protecting it. The bright layer does this for the semi-bright base — on purpose.

Semi-bright nickel: A Watts-type bath run with a sulfur-free leveler — leveled, satin, ductile, essentially sulfur-free (<0.005% S), and more noble than bright nickel. The base layer of the duplex decorative stack.

S-Rounds / sulfur-depolarized nickel: Nickel anodes with a trace of sulfur that dissolve cleanly and efficiently — the general-purpose choice; reviewed against the layer's sulfur spec on semi-bright work.

STEP test: See ASTM B764.

Substrate: The base part being plated (steel, zinc die cast, brass, etc.).

TDS (Technical Data Sheet): The supplier's instruction document for a product — always follow it for additive levels.

Throwing power: A bath's ability to plate evenly into recesses and low-current areas; improved by chloride, pH, temp, and agitation.

Triplex nickel: Duplex plus a thin high-activity nickel layer under microdiscontinuous chrome — the maximum-corrosion-service version of the stack.

Water-break test: Watching water sheet off a cleaned part — an unbroken sheet means clean; beading means oil remains.

Watts bath: The foundational nickel electrolyte (nickel sulfate + nickel chloride + boric acid), characterized by O. P. Watts in 1916 — the chassis under plain Watts, semi-bright, and bright nickel alike.



TIP

Half of “talking the language” is just these terms. When you can say “the panel's scratch lines aren't filling — leveler's probably low on the Ah schedule, let's dose per TDS and re-panel” and mean it, you're communicating like a professional plater — and your chemistry supplier can actually help you over the phone.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE MOST LIKELY STAGE, GO TO THE FULL GUIDE



REMEMBER

Most semi-bright defects trace back to **four root causes**: a prep failure upstream (cleaning/activation), a contaminated bath (metals or organics), running outside the temperature/pH/CD window, or a breach in the leveler/firewall discipline. Diagnose with a **leveling panel / Hull Cell** — read for leveling and soundness — before you start dumping chemistry.

IF YOU SEE...	MOST LIKELY STAGE TO CHECK	GO TO
Peeling / blistering of the stack	Cleaning & activation (prep)	Stations 01, 03
Skip plating / bare spots	Cleaning (buffing compound / oxide)	Station 01
Poor leveling — polish lines reproduced	Leveler low (Ah schedule)	Station 05, Ch 6.3
Dull gray patches the bright layer won't cover	Leveler breakdown products / oil drag-in	Station 05, Ch 6.3
Bright layer patchy; STEP difference collapsed	Sulfur cross-contamination (firewall breach)	Ch 6.4, Stations 05, 06
Bright bath dulling downstream	Leveler carryover through the post-plate rinse	Station 06
Pitting (gas pits)	Low wetter / poor agitation / organics	Station 05
Burning at edges/high CD	pH HIGH, temp LOW, boric low, CD high	Station 05, Ch 6.7
Dull / hazy at low CD	Metallic contamination (Cu/Zn/Pb)	Ch 6.6
Roughness / nodules	Particulate / anode fines / filtration	Station 05
Nickel bath pH dropping	Acid drag-in from activation	Station 04
Iron / metals climbing in bath	Pre-plate rinse / spent pickle	Station 04, Ch 6.6
Anodes passivating	Low chloride / anode area / non-sulfur anodes	Ch 6.5
Next layer peels off the nickel	Passivated surface — not re-activated	Station 07
Duplex fails salt spray / CASS at good thickness	STEP potentials / thickness split / sulfur pickup	Ch 3.4, Station 07
Sudden fracture of hardened part	Missed / late H-relief bake	Stations 06, 07, Ch 6.9



TIP

Notice how many symptoms trace back *upstream* of where they show up — and how the duplex-specific failures (collapsed STEP, dulling bright bath) trace back to the firewall. Most downstream defects are born upstream. Fix the root, not the symptom.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

THE TRAINER RECORDS THE DATE THE OPERATOR IS CLEARED FOR INDEPENDENT WORK AT EACH STATION



REMEMBER

An operator is “line-qualified” only when every station they run is signed. Nobody touches the **rectifier**, the **acid dip**, or **bath maintenance** solo until safety competency (PPE, SDS, LOTO, emergency response) is demonstrated. Safety qualification comes first, always.

STATION	TRAINED (DATE)	SIGN-OFF INITIALS	RE-CERT DUE
01 Soak Clean / Electroclean	_____	_____	_____
02 Rinse (Pre-Activation)	_____	_____	_____
03 Acid Activation	_____	_____	_____
04 Rinse (Pre-Plate)	_____	_____	_____
05 Semi-Bright Nickel Plating	_____	_____	_____
06 Post-Plate Rinse / Firewall / Bake	_____	_____	_____
07 Post-Treatment / Stack Completion	_____	_____	_____
08 Final Rinse / Dry	_____	_____	_____
Bath maintenance (carbon/dummy/leveler)	_____	_____	_____

Operator name: _____ Hire/start date: _____
Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



TIP

Re-verify each operator annually, and **immediately** after any process or chemistry change. Note the recertification date in the cell. People drift from procedure over time, and a yearly refresher catches bad habits before they become defects — or injuries.

• **WHERE TO GO NEXT IN THIS SERIES**

- **Bright Nickel** (*manual + poster series*) — the sacrificial mirror layer plated directly over your work: the brightener system, its own maintenance, and the other half of the firewall story.
- **Decorative Chrome** (*poster series*) — the thin top layer and its Cr(VI) safety rulebook.
- **Watts Nickel** (*manual + poster series*) — the same core chemistry with *no* additive system, run as standalone engineering nickel: build-up, salvage, electroforming. Reading it after this manual is the fastest way to understand what the leveler really does — it’s the only difference between the two baths.

20 QUESTIONS · PASSING SCORE 80% (16/20)

SEMI-BRIGHT NICKEL COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED



REMEMBER

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (Q5, Q6, Q14, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs (all must be correct):**

5, 6, 14, 19

Q1. The additive that defines a semi-bright nickel bath is:

A. A sulfur-bearing Class I carrier B. A sulfur-free leveler (coumarin or coumarin-free) C. A Class II primary brightener D. Extra nickel chloride

Q2. (Short answer) What is the **water-break test**, and what result tells you a part is clean?

Q3. In the duplex stack, which nickel layer corrodes sacrificially, and why?

A. The bright layer — its co-deposited sulfur makes it less noble than the semi-bright base B. The semi-bright layer — it is the thickest C. Neither — the chrome stops all corrosion D. Both corrode equally

Q4. (Short answer) Semi-bright is called “the layer nobody sees.” In one or two sentences, explain how the duplex stack outlasts a single nickel layer of the same total thickness.

Q5. [SAFETY GATE] Which station is described as the **single greatest source of hydrogen absorption** on the line?

A. Soak clean B. Acid activation (acid dip) C. Post-plate rinse D. Final rinse

Q6. [SAFETY GATE] A part is **> 40 HRC** (high-strength steel). Per ASTM B850, the hydrogen-embrittlement relief bake is:

A. 200°F for 1 hour, any time within a week B. 500°F for 30 minutes C. No bake — the deposit is ductile D. 375 ±25°F for at least 8 hours, started within 4 hours of the last plating step

Q7. (Short answer) The pre-activation rinse (Station 02) targets **< 500 µS/cm**, but the pre-plate rinse (Station 04) targets **< 200 µS/cm**. Why is the rinse closer to the nickel bath held to a *tighter* standard?

Q8. The semi-bright plating bath operates at what pH?

A. 3.5–4.5 B. 11.0–13.5 C. 7.0 (neutral) D. 1.5–2.2

Q9. (Short answer) What is the job of **boric acid** in the bath, and what happens if you neglect it?

Q10. The sulfur-free leveler is dosed based on:

A. The calendar — a fixed weekly addition B. The color of the bath C. Amp-hours — because it is consumed by the work, not by time D. How bright the parts look

Q11. (Short answer) A part comes off the semi-bright tank **satin, not shiny**. What should you check to judge quality at this tank — and what must you **never** add to “shine it up”?

Q12. You run a Hull Cell / leveling panel on a semi-bright bath. What are you reading it for?

- A. Brightness and brightener response B. Leveling and soundness — are scratch lines filling; any burning, roughness, pitting, contamination
C. Sulfur content D. Layer thickness

Q13. Which metallic contaminant is the **most damaging at the lowest level** (as little as 1 ppm) in the nickel bath?

- A. Iron (Fe) B. Zinc (Zn) C. Hexavalent chromium (Cr⁶⁺) D. Boric acid

Q14. (Short answer) [SAFETY GATE] Name **two** nickel-specific health hazards an operator must guard against, and one control for each.

Q15. What does the ASTM B764 **STEP test** measure?

- A. Total stack thickness only B. Salt-spray hours to failure C. The hardness of each layer D. The thickness AND electrode potential of each nickel layer individually — proving the bright layer is anodic to the semi-bright base

Q16. Dull gray patches appear on semi-bright deposits, and the bright layer won't cover them. The classic cause and fix are:

- A. Leveler (coumarin) breakdown products accumulating — carbon treat on the amp-hour schedule B. Too much boric acid — dilute the bath
C. Current density too low — double it D. Normal for semi-bright — ship it

Q17. (Short answer) Why does a **ductile, low-stress** semi-bright deposit **not** exempt a high-strength steel part from the bake?

Q18. A single drag-over of bright-nickel additive into the semi-bright tank is treated as a serious event because:

- A. It makes the parts too shiny to inspect B. It raises the bath temperature C. It adds sulfur to the base layer and collapses the STEP potential difference — killing the stack's corrosion protection, invisibly D. It only wastes a little additive — it isn't serious



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 5, 6, 14, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE you must wear at the **acid activation** station, and name the procedure you must perform before any maintenance on the **rectifier**.

Q20. (Short answer) Semi-bright nickel is never the finished surface. Name **two** honest limitations (tradeoffs) of the semi-bright process an operator should know.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY**SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD****HEADS-UP**

Keep this page out of the trainee's copy. Questions **5, 6, 14, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

- Q1. B** — A sulfur-free leveler (coumarin or a modern coumarin-free product). There is no sulfur-bearing Class I carrier and no Class II primary brightener in this bath — that package belongs to bright nickel.
- Q2.** Dip and pull the part and watch the water. If it **sheets off in an unbroken, continuous film**, the part is clean. If it **beads up**, oil remains — reclean. (It's free; use it.)
- Q3. A** — The bright layer. Its co-deposited sulfur (from the bright bath's Class I carrier) makes it **less noble** than the sulfur-free semi-bright base, so when corrosion reaches the nickel, the bright layer corrodes preferentially — sideways — protecting the base and the steel.
- Q4.** A single nickel layer lets corrosion **pit straight down** to the steel at one point. In the duplex stack the less-noble bright layer **corrodes sacrificially and spreads sideways**, so the more-noble semi-bright base — and the steel under it — stay intact. Same metal, longer life, because the stack **controls where the corrosion goes**.
- Q5. B** — Acid activation (the acid dip).
- Q6. D** — $375 \pm 25^\circ\text{F}$ for at least 8 hours, started within 4 hours of the last plating step (ASTM B850). (Bonus concept: on a duplex part the "last plating step" is usually the bright nickel or chrome — verify per the customer spec.)
- Q7.** Because tighter rinsing closer to the nickel bath limits **drag-in** of acid, dissolved **iron**, and other metals. The bath is **maintained for years**, so contaminants **accumulate permanently** and are hard to remove — the closer to the tank, the more damage drag-in does, so the standard tightens. (Full credit for naming pH protection, iron control, or "permanent accumulation.")
- Q8. A** — 3.5–4.5.
- Q9.** Boric acid is the **pH buffer** — it holds pH steady right at the part's surface (the cathode film) during plating. Neglect it and you get **burning** at high-current areas and inconsistent deposits.
- Q10. C** — Amp-hours. The leveler is consumed by the plating work itself, so additions track the rectifier's Ah totals per the supplier's TDS — never the calendar, never by eye.
- Q11.** Check **leveling** — are polish lines and scratches filling (leveling panel / Hull Cell), and is the satin sound and uniform? Satin is *correct* for semi-bright; the mirror comes from the bright tank downstream. **Never add bright-nickel (sulfur-bearing) brightener** to this tank — it would put sulfur in the deposit and destroy the layer's nobility.
- Q12. B** — Leveling and soundness (scratch fill, burning, roughness, pitting, contamination). There's no brightener to read for.
- Q13. C** — Hexavalent chromium (Cr^{6+}), damaging at ~1 ppm — and on a duplex line the chrome tank is just downstream, so drag-back is a live threat.
- Q14.** Any two, each with a control: **skin sensitization / "nickel itch"** → gloves at every nickel station, minimize skin contact; **nickel mist (suspected carcinogen)**, **PEL 1 mg/m³ as Ni** → local exhaust ventilation, respirator when ventilation is inadequate; **ingestion** → no eating/drinking/smoking on the line, wash before breaks. (Any two valid hazard + control pairs.)
- Q15. D** — Each nickel layer's thickness AND electrode potential, measured individually as the layers are anodically dissolved — confirming the bright layer is anodic (less noble) to the semi-bright base. Right thicknesses with wrong potentials still fails.
- Q16. A** — Accumulated leveler (coumarin) breakdown products; the fix is carbon treatment on the set amp-hour schedule (then re-dose the leveler and confirm by panel).
- Q17.** Because the ductility is a property of the **nickel deposit**, but the embrittlement is in the **steel substrate** underneath — from hydrogen absorbed during cleaning, activation, and plating. The bake drives that hydrogen out of the steel; the deposit being ductile doesn't change the steel's condition.
- Q18. C** — Bright-bath additive carries sulfur; sulfur in the semi-bright layer collapses the STEP potential difference and kills the duplex corrosion mechanism — and the parts *look completely normal*, so the failure is invisible until a STEP report or a field return. That's why the firewall rules are absolute.
- Q19. PPE (any four):** sealed chemical splash goggles (*not* safety glasses on this acid station), face shield, acid-resistant gloves, chemical apron, chemical-resistant boots, respirator if ventilation is inadequate. **Rectifier procedure: LOTO (Lockout/Tagout)** — de-energize and lock out the power before any rectifier maintenance.
- Q20.** Any two of: **not decorative as-plated** (satin — needs bright + chrome over it to be a finished surface); **the leveler must be managed** (dosed on Ah, breaks down, needs scheduled carbon treatment); **two nickel chemistries on one line** demand firewall discipline (cross-contamination is a classic failure); **sensitive to metallic contamination** (Cu/Fe/Zn/Cr accumulate in a maintained bath); (also acceptable: runs hot, so energy cost; H-embrittlement bake required on hardened steel).

MC letter distribution (Q1, 3, 5, 6, 8, 10, 12, 13, 15, 16, 18): A=3 · B=3 · C=3 · D=2 (11 multiple-choice items; 9 short-answer). Safety short-answers (Q14, Q19) are graded on concepts, not letters.

**REMEMBER**

A score of 16/20 passes — but every safety question (5, 6, 14, 19) must be correct to certify. If a trainee misses a safety item, re-teach it and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN



PLATING POSTERS

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CERTIFICATE OF COMPLETION
SEMI-BRIGHT NICKEL PLATING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Semi-Bright Nickel Plating** training course — covering all eight process stages (Clean, Rinse, Activate, Rinse, Plate, Rinse, Post-Treat, and Final Rinse/Dry), the duplex/multilayer nickel system and its STEP verification, the semi-bright bath chemistry and its sulfur-free leveler system, the semi-bright/bright firewall discipline, anode selection, contamination control, leveling-panel diagnostics, internal stress and ductility, hydrogen-embrittlement control, troubleshooting, and the safety practices for alkaline cleaners, mineral acids, the acid nickel bath, and post-treatment — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Technical training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Nickel compounds are classified as suspected carcinogens and known skin sensitizers; hexavalent chromium, where used in a topcoat, is a confirmed human carcinogen — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations. *Regulatory data reviewed 2026-09-18: exposure limits reflect published U.S. federal OSHA PELs as of this date (soluble nickel 1 mg/m³ TWA as Ni; sulfuric acid mist 1 mg/m³ TWA; HCl 5 ppm ceiling; Cr(VI) 5 µg/m³ TWA, action level 2.5 µg/m³). Referenced standards (ASTM B456, B764, B571, B850, F519; ISO 1456; AMS 2403) should be checked against their current revision. State limits (e.g., Cal/OSHA) and ACGIH TLVs may be stricter. Boric acid: treated as a reproductive-hazard concern (no federal OSHA PEL in the confirmed set). Verify against current OSHA tables and supplier SDS before production use.*

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